



Review

On the role of linguistic features for comprehension and learning from STEM texts. A meta-analysis

Anselm R. Strohmaier^{a,b,*}, Timo Ehmke^{b,c}, Hendrik Härtig^{b,d}, Dominik Leiss^{b,e}

^a Technical University of Munich, TUM School of Social Sciences and Technology, Department of Educational Sciences, Munich, Germany

^b Leuphana University of Lüneburg, Center for Empirical Research on Language and Education, Lüneburg, Germany

^c Leuphana University of Lüneburg, Institute of Educational Sciences, Lüneburg, Germany

^d University of Duisburg-Essen, Department of Physics Education, Essen, Germany

^e Leuphana University of Lüneburg, Institute of Mathematics and its Didactics, Lüneburg, Germany



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ABSTRACT

Learning from texts is an important part of STEM education, but linguistic text features can affect comprehension and learning. We synthesized 45 experimental studies that investigated the effect of linguistic modifications on STEM text comprehension and learning, with a total of $N = 6477$ learners. In a meta-regression, we distinguished between eight categories of linguistic modifications and several other moderators, including prior knowledge. We found a small positive mean effect of linguistic modifications ($g = 0.15$). In line with previous research, the results showed that learners with low content prior knowledge benefited more from linguistic modifications. Personalization and increasing clarity and elaboration had positive effects, while reducing complexity and increasing cohesion did not have significant effects. This study highlights the potential and limits of linguistic modifications in STEM texts and provides implications for educational practice.

1. Introduction

Texts are important in Science, Technology, Engineering, and Mathematics (STEM)¹ education in schools and universities (Schlepppegrell, 2007; van den Broek, 2010). Comprehending texts is a complex cognitive task that is influenced by the linguistic features of the text and various aspects of the reader (McNamara & Magliano, 2009). STEM texts in particular present content-related challenges but also unique linguistic features that make them difficult to understand for many readers (Best et al., 2005; Cruz Neri & Retelsdorf, 2022; Goldman & Bisanz, 2002; van den Broek et al., 2002). To design good texts for STEM education, it is important to understand how changes to linguistic features, which we refer to as *linguistic modifications*, can positively impact comprehension and learning for particular groups of learners.

Previous research has addressed this question by systematically investigating effects of linguistic modifications of STEM texts on comprehension and learning outcomes. However, results from these studies are somewhat inconclusive: While some studies report

* Corresponding author. Technical University of Munich, TUM School of Social Sciences and Technology, Department of Educational Sciences, Arcisstr. 21, 80333, Munich, Germany.

E-mail address: anselm.strohmaier@tum.de (A.R. Strohmaier).

¹ Some research on text comprehension simply refers to *science texts*. Graesser et al. (2002) point out that this largely coincides with the term STEM, because it typically also relates to the domains of Technology, Engineering, and Mathematics.

substantial effects, others do not. Various different approaches of linguistic modifications were used, focusing on different linguistic features in different combinations. Moreover, it has been shown that individual characteristics interact with linguistic features, and that therefore, a linguistic modification might be helpful for one group of learners, but not for the other (Best et al., 2005; McNamara et al., 1996). This makes it difficult for researchers and practitioners to decide whether and how to linguistically modify STEM texts for their specific purpose (Cruz Neri & Retelsdorf, 2022).

In the present study, we addressed this issue by conducting a meta-analysis on the effects of linguistic modifications on comprehension and learning from STEM texts. By simultaneously including several linguistic modifications in a meta-regressional approach, we isolated the unique contributions of each modification. Moreover, we included various characteristics of the studies and their samples in order to identify factors that moderated the effectiveness of linguistic modifications.

This study aims to provide guidance for researchers and educators from STEM disciplines to construct, analyze and modify texts for optimal learning and comprehension. This goal reflects the long-standing call to summarize research findings from research on text comprehension in a way that allows teachers from non-linguistic disciplines to implement them in their classrooms in order to provide their students with optimal support and appropriate assessment (e.g., Fang et al., 2008; Hand et al., 2010; Snow, 2002).

In the following sections, we provide an overview of the process of text comprehension and learning from texts and outline the specific characteristics and challenges of reading STEM texts. We review studies that aimed to improve STEM texts for comprehension and learning, suggesting a categorization of linguistic modifications. Additionally, we review the role of prior knowledge before summarizing open questions and outlining the goals of this study.

1.1. Comprehension and learning from texts

1.1.1. Models of text comprehension

Text comprehension is the process of constructing a *mental representation* of a text, capturing its elements and their semantic relations (Goldman & Bisanz, 2002; van den Broek et al., 2002; see McNamara & Magliano, 2009, for a review). The terms *situation model* (van Dijk & Kintsch, 1983) or *knowledge structure* (Follmer et al., 2018) describe concepts similar to mental representations, but with nuanced differences.

The process of building the mental representation involves (1) interpreting the meaning of words and larger linguistic units of the text, (2) determining their semantic connections, and (3) relating it to other information, such as prior knowledge or preceding sentences (van den Broek et al., 2002). The resulting mental representation is unique to each reader (Schnitz & Bannert, 2003) and contains information that is directly retrieved from the text, but also information that is added by the reader (McNamara & Magliano, 2009). Therefore, individual characteristics and cognitive resources will influence the processes and the products of text comprehension (Cromley & Azevedo, 2007; Retelsdorf et al., 2011).

The decoding of words and the semantic *mapping* of elements of the text with each other and with prior knowledge is mostly automatic, but conscious inferences may be necessary if automatic mapping is not successful. This requires the availability of cognitive resources such as working memory capacity (Bohn-Gettler & Kendeou, 2014; Kintsch & van Dijk, 1978) and executive functions (Butterfuss & Kendeou, 2017; Follmer & Sperling, 2018) and can be related to Cognitive Load Theory (Ozuru et al., 2009).

1.1.2. Comprehension and learning

In the present study, we refer to *learning* as the “incorporation of comprehended information into the reader’s background knowledge” (van den Broek, 2010, p. 453). This narrow definition is limited to the content of the text as the learning object. In reality, students can arguably learn more from a STEM text than just its content, which is not the focus of this paper.

Comprehension and learning are intertwined during the reading of STEM texts. In theory, text comprehension finishes with a mental representation, and learning (in this narrow sense) begins when comprehended information is integrated with the readers’ knowledge base. Thus, per definition, comprehension precedes learning. However, it is practically impossible to distinguish between the two: On the one hand, the process of text comprehension constantly relates prior knowledge and information from the text and has the potential to thereby expand and modify the reader’s knowledge base (McNamara & Magliano, 2009). On the other hand, the process of learning will include revisiting aspects of the text and comprehend it again. Therefore, we use the term *comprehension and learning* in the current paper and refer to all aspects of the construction of a mental representation from a text and its integration with the reader’s knowledge base.

Within this terminology, we distinguish three levels of comprehension and learning based on the tasks that are used for assessment: *recall*, *application*, and *transfer* (McNamara & Magliano, 2009). Recall tasks assess the ability to reproduce explicit textual information. They require superficial comprehension and some learning in the form of memorization, but no integration with prior knowledge. Application tasks require the reader to process the comprehended information, for example, by making inferences, which makes it more likely that the content of the text has been integrated with prior knowledge. Finally, transfer tasks require the integration of the comprehended information with the readers knowledge base, which indicates deeper comprehension and learning.

1.1.3. Comprehension and learning from STEM texts

STEM texts present specific challenges in text comprehension compared to other types of texts, which can be structured according to the three processes of building a mental representation (1) interpreting elements of the text (2) determining semantic connections across elements, and (3) relating it with prior knowledge (see Sect. 1.1.1; van den Broek et al., 2002).

- (1) STEM texts are particularly challenging regarding word decoding and concept building, even for proficient readers (Deese, 1962; Fang et al., 2006; Follmer et al., 2018). Process models of text comprehension typically focus less on lower-level processes like word decoding (McNamara & Magliano, 2009, but see Cromley et al., 2010, for a counterexample) but STEM texts frequently use technical vocabulary and academic language, which are more difficult to decode, trigger fewer background knowledge and spread less cognitive activation (Best et al., 2005; Graesser et al., 2011; Schleppegrell, 2001; Snow & Uccelli, 2009). In addition, STEM texts often have a complex syntax, which makes it more difficult to parse sentences and extract meaning, and adds demands on readers' working memory by increasing the information required to be processed simultaneously (Best et al., 2005; Graesser et al., 2011).
- (2) Determining the semantic connections between elements of the text is particularly challenging for STEM texts because they are typically informational or expository texts with a logical semantic structure, which is harder to comprehend than the linear temporal structure of basic narrative texts (Coté et al., 1998; Follmer et al., 2018; Graesser et al., 2002; van den Broek, 2010; Weaver & Kintsch, 1991). This makes it more difficult to establish semantic links and construct mental representations of STEM texts.
- (3) STEM texts are challenging in terms of inferring the relation between the text and prior knowledge (Ozuru et al., 2009). Their purpose in educational settings is often to teach complex, new knowledge and unknown concepts or to assess such knowledge, which is often unfamiliar, difficult to map, and may even conflict with prior knowledge (van den Broek, 2010; Tippet, 2010; Weaver & Kintsch, 1991). This makes STEM texts notoriously hard to understand as text comprehension heavily relies on activating and using prior knowledge to build inferences (Best et al., 2005; Cromley et al., 2010; Follmer et al., 2018).

1.1.4. STEM text comprehension and cognitive load theory

Because working memory is an important resource in reading, *Cognitive Load Theory* can be used to describe how linguistic features and characteristics of the reader can influence comprehension and learning from STEM texts (Paas et al., 2003; Sweller, 1994; Sweller et al., 2019; see also Kintsch & van Dijk, 1978). Cognitive Load Theory divides cognitive load into three categories: *intrinsic*, *extraneous*, and *germane*. These categories compete for limited working memory capacity².

In reading STEM texts, intrinsic load is related to the inherent complexity of the STEM content and is lower for students with higher content prior knowledge (Sweller et al., 2019; van Merriënboer & Sweller, 2005). Extraneous load refers to cognitive processes that are irrelevant for successful comprehension and learning. Among other factors, it can potentially be influenced by the design of the text: Linguistic complexity of STEM texts can be considered extraneous load because it requires working memory capacity but does not, by itself, contribute to the building of a mental representation (Mikk, 2008). Germane load is the necessary and desirable cognitive load for comprehension and learning.

All three aspects of cognitive load are relevant for supporting comprehension and learning from STEM texts: If prior knowledge is higher, intrinsic load might be reduced, supporting knowledge construction (Sweller et al., 2019; van Merriënboer & Sweller, 2005). By reducing extraneous load, cognitive resources for germane load are freed up (Mikk, 2008) and effective learning environment design and promoting effective reading strategies can further enhance germane load (Leutner et al., 2009; Sweller et al., 1998).

1.2. Improving STEM texts for learning and comprehension with linguistic modifications

Previous research has aimed to improve comprehension and learning from STEM texts by modifying linguistic features which targeted one or more of the three steps in the process of building a mental representation (see Sect. 1.1.1). Accordingly, we propose to classify linguistic modifications in three categories: The first category are modifications which support the reader in interpreting the meaning of text elements by *reducing linguistic complexity*. This should make the process of decoding easier (McNamara & Magliano, 2009) and reduce extraneous cognitive load (Mikk, 2008). The second category are modifications that make semantic links of elements within a text more explicit (i.e., *increasing cohesion*³). This aims at making it easier for the reader to establish a connected mental representation and decreasing the amount of information that has to be kept in the working memory, also decreasing cognitive load (van den Broek, 2010). The third category are modifications that support the reader in integrating prior knowledge with the information from the text by *increasing clarity and elaborations*. These three categories may overlap; for example, more common words are easier to decode but will also spread activation more strongly, which helps the reader to infer semantic links (Graesser et al., 2011). Conversely, the categories may also contradict each other: Connecting sentences to increase cohesion simultaneously increases syntactic complexity.

In line with previous research (e.g., Cruz Neri & Retelsdorf, 2022; McNamara & Magliano, 2009), we propose to further distinguish between *local* linguistic modifications like words or propositions within a sentence and *global* linguistic modifications which refer to the text and its content as a whole.

This two-dimensional distinction leads to a 2×3 matrix of categories of linguistic modifications which is given in Table 1. In the following, we summarize the linguistic modifications that were investigated by existing studies based on this categorization.

² In a more recent revision of Cognitive Load Theory (Sweller et al., 2019), germane load is no longer considered to add to the total load, but rather refers to the redistribution of resources from extraneous to intrinsic load.

³ The term *cohesion* is sometimes used as an umbrella term for different linguistic features of a text, also including links with prior knowledge (e.g., Follmer & Sperling, 2018; Graesser et al., 2011). In this paper, we refer to a cohesion in a narrower sense of *situational* and *referential cohesion* as described by McNamara and Magliano (2009).

Table 1
Categories of linguistic modifications.

Level	Linguistic modification		
	Reducing linguistic complexity	Increasing links within the text	Increasing links with prior knowledge
Local	Reducing word-level complexity	Increasing local cohesion	Increasing local clarity and elaboration
Global	Reducing syntactic complexity	Increasing global cohesion	Increasing global clarity and elaboration

1.2.1. Reducing word-level complexity

Word-level complexity was decreased by replacing less frequent words with more frequent words (Arya et al., 2011; Plath, 2020; Wenger & Spyridakis, 1993), longer words with shorter words, or technical vocabulary with more general terms (Burton, 2014). Because technical vocabulary might be part of the learning content, other studies only simplified non-technical terms (Härtig et al., 2019; Williams, 1968). Complex grammatical structures on the word level like compounds, nominalizations, appositives or derivatives were also avoided (Arya et al., 2011; Härtig et al., 2019; Plath, 2020; Wenger & Spyridakis, 1993).

These studies did not demonstrate clear unique effects of decreasing word-level complexity on comprehension and learning, since only two studies modified word-level complexity without, at the same time, also varying other linguistic aspects. These studies reported mixed results (Arya et al., 2011; Burton, 2014).

1.2.2. Reducing syntactic complexity

Syntactic complexity was reduced by shortening sentences (Burton, 2014; Hacker et al., 1992; Ikeda & Kitagami, 2013; Walkington et al., 2017), splitting up subordinate and main clauses (Hacker et al., 1992; Härtig et al., 2019; Wenger & Spyridakis, 1993), or by simplifying the syntax (Arya et al., 2011; Meneses et al., 2018; Plath, 2020; Schneider et al., 2018). Two studies reduced the use of passive voice (Plath, 2020; Wenger & Spyridakis, 1993). Only Walkington et al. (2017) analyzed the effect of syntactic complexity in isolation and did not find a significant effect on comprehension.

1.2.3. Increasing local cohesion

Cohesion refers to the clarity of semantic, referential, causal, temporal or spatial ties of elements within a text. *Local cohesion* refers to the connection within and between consecutive sentences and sentences in close proximity (McNamara & Magliano, 2009, van Dijk & Kintsch, 1983). Local cohesion is typically improved either by increasing referential overlap between sentences or by signaling the connections between sentences with connectives.

Increasing referential overlap was achieved by using of the same terms repeatedly instead of replacing them with synonyms (Boscolo & Mason, 2003; Deppert, 2001; Follmer & Sperling, 2018; Hall et al., 2016; Härtig et al., 2019; Meneses et al., 2018; Schneider et al., 2018). Similarly, avoiding unspecific pronouns and repeating explicit nouns is assumed to increase cohesion (Degand et al., 1999; Gasparinatou & Grigoriadou, 2010, 2011, 2013; Hall et al., 2014; Hall et al., 2015; Ikeda & Kitagami, 2013; Kamalski et al., 2008; McNamara, 2001; McNamara et al., 2011; O'Reilly & McNamara, 2007; Ozuru et al., 2010; Ozuru et al., 2009; Wittwer & Ihme, 2014). The overlap between sentences was increased by using similar phrases and words for similar concepts across the text (Ikeda & Kitagami, 2013; McNamara, 2001; McNamara et al., 2011; O'Reilly & McNamara, 2007; Ozuru et al., 2010; Ozuru et al., 2009). No studies investigated the effect of increasing referential overlap in isolation, so the individual effect of this modification is unclear.

Local cohesion was also improved by using *connectives*, which are words or phrases that serve the function of logically connecting sentences or clauses (e.g., Degand et al., 1999). Most studies compared the use of connectives with a control text without connectives. A majority of studies did that in connection with other modifications (Boscolo & Mason, 2003; Deppert, 2001; Gasparinatou & Grigoriadou, 2010, 2011, 2013; Hall et al., 2014; Hall et al., 2015; Ikeda & Kitagami, 2013; Kamalski et al., 2008; McNamara, 2001; McNamara et al., 1996; McNamara et al., 2011; Meyer et al., 1980; O'Reilly & McNamara, 2007; Ozuru et al., 2010; Ozuru et al., 2009; Vidal-Abarca & Sanjose, 1998), but some studies focused exclusively on manipulating connectives. Of these studies, Degand et al. (1999), Degand and Sanders (2002), Maury and Teisserenc (2005), and Spyridakis and Standal (1987) reported positive effects. In contrast, Hall et al. (2016) did not find a significant effect, Geva and Ryan (1985) reported mixed results for different groups of learners, and Millis et al. (1993) reported negative effects. These conflicting results are in line with a theoretical debate which contrasts positive effects (*connective facilitation*) and negative effects (*connective interference*) of connectives (Degand et al., 1999).

1.2.4. Increasing global cohesion

Corresponding to local cohesion, *global cohesion* refers to the clarity of the connection between each part of the text with the text as a whole (McNamara & Magliano, 2009). Studies aimed at increasing global cohesion by restructuring the text, for example, in a chronological order (Härtig et al., 2019; McNamara et al., 2011), to better address the reader's interest and prior knowledge (Yaros, 2006, 2011), in a way so that sentences appear in thematical proximity (McNamara & Magliano, 2009; O'Reilly & McNamara, 2007), in a logically more linear order (Deppert, 2001; Schneider et al., 2018; Vidal-Abarca & Sanjose, 1998), or in line with a cause-effect structure (Snyder, 2012). No studies investigated these manipulations in isolation.

1.2.5. Increasing local clarity and elaboration

In the context of text comprehension, *elaboration* primarily refers to making inferences between the text and prior knowledge as

well as other parts of the text (Graesser et al., 2002; McNamara & Magliano, 2009). Elaborations can be made by the reader, but they can also be made explicit in the text to increase clarity.

Several studies modified the clarity of STEM texts and added elaborations locally, that is, within the sentence. For example, some texts repeated or added important information in order to make it more explicit (Meyer et al., 1980; Ortegren et al., 2015; Snyder, 2012; Vidal-Abarca & Sanjose, 1998). Other studies explained difficult words (Deppert, 2001; Härtig et al., 2019; Ikeda & Kitagami, 2013; Meneses et al., 2018; Williams, 1968; Yaros, 2006, 2011) and unfamiliar concepts (Hall et al., 2014, 2015; Meneses et al., 2018). In some studies, connections between elements of the text and prior knowledge were made more explicit (Gasparinatou & Grigoriadou, 2010, 2011, 2013; Hacker et al., 1992; Härtig et al., 2019; Ozuru et al., 2010; Schneider et al., 2018; Snyder, 2012; Vidal-Abarca & Sanjose, 1998; Wittwer & Ihme, 2014). Several studies added *descriptive elaborations* (Ikeda & Kitagami, 2013; McNamara, 2001; McNamara et al., 2011; O'Reilly & McNamara, 2007; Ozuru et al., 2009), for example, by adding explanatory insertions.

Only three studies reported isolated effects of elaborations: Wittwer and Ihme (2014) reported a significant positive effect of specifying causal relations on comprehension. Ortegren et al. (2015) reported that the repetition of information significantly increased text recall. Vidal-Abarca and Sanjose (1998) found differing results for different outcome measures.

1.2.6. Increasing global clarity and elaboration

Apart from elaborations within a sentence, elaborations and clarity can also be increased on a global text level. This was done by including summarizing headings and subheadings, previews, thematic sentences, and macropropositions which direct the attention to the core ideas from the text (Boscolo & Mason, 2003; Gasparinatou & Grigoriadou, 2010, 2011; Kamalski et al., 2008; McNamara, 2001; McNamara et al., 1996; McNamara et al., 2011; O'Reilly & McNamara, 2007; Ozuru et al., 2009; Schneider et al., 2018; Spyridakis & Standal, 1987; Wenger & Spyridakis, 1993). Other studies made headings more concrete (Lippmann et al., 2019), added summaries that repeated core information from the text (Vidal-Abarca & Sanjose, 1998), added advance organizers or organizational signals that elaborated on the text structure (Gasparinatou & Grigoriadou, 2010; Kamalski et al., 2008; Sánchez et al., 2017; Wenger & Spyridakis, 1993) or explicitly stated a top-level structure (Meyer et al., 1980). Studies which investigated such modifications in isolation reported mixed results, including substantial differences based on prior knowledge (Boscolo & Mason, 2003; Gasparinatou & Grigoriadou, 2010; McNamara et al., 1996), inconclusive results (Gasparinatou & Grigoriadou, 2011; Spyridakis & Standal, 1987) or no effects (Lippmann et al., 2019).

Personalization aims to increase readers' elaboration of the text by increasing involvement, motivation, and engagement, for example, by using second person personal pronouns and possessive pronouns instead of definite articles (Cordova & Lepper, 1996; Ginns et al., 2013). Three studies investigated the effect of personalization on learning from STEM texts (Dutke et al., 2016; Moreno & Mayer, 2000; Stiller & Jedlicka, 2010), which all addressed the effect in isolation. Dutke et al. (2016) and Moreno and Mayer (2000) both found substantial effects for transfer, but not for application outcomes. Stiller and Jedlicka (2010) reported positive effects only for learners with low prior knowledge.

1.3. Prior knowledge and linguistic modifications

Apart from investigating the effects of linguistic modifications, previous research on text comprehension has tried to identify moderating characteristics of readers that determine if such modifications affect comprehension and learning or not (e.g., McNamara et al., 1996; McNamara & Kintsch, 1996). Because processes of comprehension and learning largely involve an interaction between the text and the reader (Cromley & Azevedo, 2007), individual differences are likely to occur (Best et al., 2005). Thus, the optimal text modification might depend on the prior knowledge of the reader, both with regard to the content domain and to language (van den Broek, 2010).

1.3.1. Content prior knowledge

Various studies by McNamara and colleagues found that increasing text cohesion helps students with lower prior knowledge, but is substantially less beneficial for high prior knowledge students (e.g., McNamara, 2001; McNamara et al., 1996; McNamara & Kintsch, 1996; McNamara et al., 2011; Ozuru et al., 2009, but see also O'Reilly & McNamara, 2007, for opposite findings). This is known as the *reverse cohesion effect* (McNamara et al., 1996; O'Reilly & McNamara, 2007). More generally, this phenomenon can be understood as a specific case of an *expertise reversal effect* (Kalyuga et al., 2003; Ozuru et al., 2009) and could also affect linguistic modifications other than increasing cohesion. Possibly, linguistic modifications might help some readers to tackle difficult texts, but could hinder self-directed cognitive activities for others (Wittwer & Renkl, 2008).

Few studies have specifically addressed the expertise reversal effect beyond cohesion or particularly from a perspective of learning from STEM texts (Best et al., 2005; Follmer & Sperling, 2018). The effect might be stronger in STEM texts as content prior knowledge is important for comprehension, and students differ greatly in prior STEM knowledge (Alexander et al., 1994; Graesser et al., 2002). Low prior knowledge students might benefit from linguistic modifications to compensate for their disadvantage, while high prior knowledge students might particularly benefit from the challenge to activate, use, and connect their knowledge themselves.

1.3.2. Language-specific prior knowledge

The reverse cohesion effect was first described for content prior knowledge. However, it seems also plausible that similar effects exist for linguistic abilities, or language prior knowledge. Similar to content prior knowledge, readers with high language prior knowledge might not benefit from linguistic modifications. However, it is also possible that their language prior knowledge enables them to take advantage of modifications (Ozuru et al., 2009). Findings on this effect are somewhat unclear: Geva and Ryan (1985)

report a language-specific expertise reversal effect for some, but not all of their linguistic modifications. In contrast, [Ozuru et al. \(2009\)](#) and [O'Reilly and McNamara \(2007\)](#) found that some participants with high language prior knowledge gained more from a high cohesion text.

2. The present study

Previous research has found that modifying linguistic features of STEM texts can impact comprehension and learning. However, the results of these studies are inconsistent and make it difficult for researchers and practitioners to determine effective modifications. This is due to:

(1) Studies usually addressing multiple modifications at once, with few studies utilizing linguistic modifications in isolation, making it unclear which particular modification was effective; (2) the use of different texts from study to study, so that the same linguistic modification might be differently effective in one text or the other ([Follmer et al., 2018](#)); (3) small sample sizes in some studies, leading to many inconclusive results. (4) the potentially differential effectiveness for different groups of learners (e.g., [McNamara et al., 1996](#)) which limits transferability of results.

These difficulties lead to a situation where substantial evidence has been gathered throughout recent decades, but still without a clear conclusion on which particular linguistic modifications improve learning with STEM texts, and for which group of learners. In the present study, we approached this gap by combining existing studies in a meta-analytical approach to addressing the following research questions.

- (1) What is the overall effect of linguistic modifications on comprehension and learning from STEM texts?
- (2) How do content prior knowledge and language prior knowledge moderate the effect of linguistic modifications on comprehension and learning from STEM texts?
- (3) Which categories of linguistic modifications (reducing complexity, increasing internal links, and increasing links with prior knowledge) at both local and global levels improve comprehension and learning from STEM texts?
- (4) Which other study characteristics (assessment outcome, publication year, domain, text length, text language, age group, study location) moderate the effect of linguistic modifications on comprehension and learning from STEM texts?

3. Methods

3.1. Study selection

Relevant studies were identified by database searches and cross-reference checks. Database searches were conducted in *Web of Science*, *Scopus*, *PSYINDEX*, *ProQuest* and *FIS Bildung*, including studies published up to and including 2020. Approximately 50 search terms referring to *text characteristics*, *outcome*, *STEM* and *material* were used (see [Table A.1](#)).

The database search resulted in 1525 hits (excluding duplicates). Abstracts were screened independently by a subset of two of the four authors. If they disagreed, an independent screening was done by a third author based on the same criteria. The inclusion criteria were.

- 1) The study used a linguistic modification of a text. We did not include studies that focused only on variations of the content or structural level. For example, we did not include studies implementing a refutational structure (see [Tippett, 2010](#), for an overview), or studies investigating the effect of seductive details (see [Sundararajan & Adesope, 2020](#), for a meta-analysis), since the mechanisms behind these modifications are arguably not based primarily on linguistic processes.
- 2) The study included an experimental design with at least two groups or conditions regarding linguistic modifications.
- 3) The study reported a measure of text comprehension or achievement as an outcome variable, and this measure was the same for all experimental groups.
- 4) The manipulated text covered STEM or science topics (based on the description of these domains in [Graesser et al., 2002](#)).
- 5) The study used linear, written texts (no hypertexts or websites). This could include digital texts, but only if the structure was similar to printed texts.
- 6) The study was written in English or German, but the texts used could be in any language.

Based on this screening, 77 studies were categorized as possibly relevant by both screeners, 1364 were categorized as irrelevant. 84 cases resulted in a disagreement (70 of them regarding Criterion 1), in 31 of these cases the third coder decided to include the studies. This resulted in a total of 108 possibly relevant studies, which were retrieved as full texts. These full texts were again screened based on the same criteria by the first author and the inclusion of each study was discussed by all four authors. In contrast to the first screening, this also included information from the full text, including details of the design and topic of the texts, the outcome measures, and the statistical design. This led to the exclusion of 70 studies. Finally, 38 studies were selected and 9 additional studies were included through cross-reference searches within all screened full texts. In cases where the reported information was not sufficient for coding, authors of the studies were contacted. In two cases, the missing information could not be retrieved and the studies were excluded from the dataset. Ultimately, 45 studies were included for analyses.

3.2. Coding of moderators

The coding process closely followed the suggestions by [Lipsey and Wilson \(2000\)](#). A detailed coding sheet was created for the coding process. For the coding of linguistic modifications, the process included an explorative phase and a validation phase with double-coding by two authors (see Section 5.2.3). All other variables were coded only by the first author.

3.2.1. Linguistic modifications

Initial coding of linguistic modifications was based on the 2×3 matrix described in Section 2. During the explorative coding phase, two of the six categories were split, resulting in a total of eight categories: The category *local cohesion* was split into *overlap* and *connectives*. The category *increasing global clarity and elaboration* was split into *summaries* and *personalization*.

Based on the resulting eight categories a detailed coding scheme was created (see [Table A.2](#)). Each linguistic modification was dummy coded in all eight categories (*yes* = 1, *no* = 0).⁴ Two authors then individually coded all linguistic modifications according to the coding scheme, resulting in an agreement of $\kappa = 0.84$. Consensus was reached in a discussion between both coders.

3.2.2. Prior knowledge

Language-specific and content prior knowledge were categorized as *low* (= -1), *regular* (= 0), or *high* (= 1), respectively. The descriptions of samples in the original studies were used as criteria for this categorization. If a study distinguished between two groups by splitting it based on language-specific or content prior knowledge, they were coded as *high* and *low*; if there was only one group, it was coded as *regular*, if not specifically characterized otherwise (e.g., in [Gasparinatou & Grigoriadou, 2011](#)).

3.2.3. Study descriptors

To characterize the level of comprehension and learning that studies assessed as outcome measures, we distinguished assessment outcomes in three categories: *Recall* measures referred to a verbatim reproduction of text-based information (e.g., number of correctly recalled propositions in a free recall task, [Boscolo & Mason, 2003](#); or forced-choice fact retrieval questions, [Wenger & Spyridakis, 1993](#)), *apply* measures referred to questions that could be answered by combining information that was fully given in the text (e.g., bridging-inference questions where information from two separate sentences had to be combined, [Gasparinatou & Grigoriadou, 2010](#)), *transfer* measures referred to questions which required information that was not given directly in the text and had to be inferred or taken from prior knowledge (e.g., problem-solving questions, [Stiller & Jedlicka, 2010](#); [Vidal-Abarca & Sanjose, 1998](#); [Wenger & Spyridakis, 1993](#)). For 172 extracted effects (91.5%) one of the three categories could be distinctly assigned (78 recall, 66 apply, 28 transfer), while the remaining 16 effects were based on a combination of the assessment types. 28 of the 45 studies (62%) reported effects for outcome measures in more than one category.

Domain was coded by the content of the text as self-reported in the studies. Studies that used texts from different scientific domains, science-overarching topics or referred to themselves simply as *science text* were categorized as *mixed science* (14 studies). The domains geography, mathematics, and psychology were combined to *other* because of the limited number of studies in each domain (2, 2, and 1, resp.). Physics (6 studies), computer science (3 studies), and biology (18 studies) were coded as individual domains.⁵ *Text length* was coded by number of words. If the number of pages was the only available information, 450 words per page were assumed. Due to the small number of studies using texts in other languages, we only distinguished between text in *English*, *German*, and *other languages*. Age groups of participants were distinguished by grade level, age was used if no grade level was reported. We distinguished between *primary education* (grades 1–6, ages 5–11), *lower secondary education* (grades 7–10, ages 12–15), *higher secondary education* (from grade 11, from age 16). No studies investigated pre-schoolers, all participants beyond secondary education were categorized as *post-secondary*. For *study location*, we distinguished between studies that were conducted in a laboratory setting other than the regular learning environment, in the regular learning environment but in an irregular setting (e.g., individual testing at school), during regular classroom activities, or in an online assessment.

3.2.4. Extraction of effect sizes

The unbiased standardized mean difference (Hedges' *g*) was used as dependent variable ([Hedges, 1981](#)). Effect sizes were obtained following the methods described in [Borenstein et al. \(2009\)](#) and [Lipsey and Wilson \(2000\)](#).

Effect sizes were taken directly from the studies or, if this was not possible, re-calculated from reported mean values, standard deviations and sample sizes of the relevant groups. If necessary, Hedges' *g* was derived from other effect size measures according to the procedures reported in [Borenstein et al. \(2009\)](#) and [Lipsey and Wilson \(2000\)](#).

Error variances of effect sizes were estimated proportionally to the sample size ([Borenstein et al., 2009](#)). If necessary, error variances were adjusted for nested designs: In repeated-measures designs, the correlation between two measurement occasions was estimated as $r = .30$ if it was not explicitly reported, based on reported correlations in [Hall et al. \(2014\)](#) and [Hall et al. \(2016\)](#). In cases where results from different tasks were combined (e.g., when two tests of the same outcome were used but reported separately) and no

⁴ Of the 65 different linguistic modifications, 36 affected only one category, nine affected two categories, nine affected three categories, five affected four categories, and six affected five categories at the same time.

⁵ One study included a physics text and a geography text.

correlation was reported, a correlation of $r = .80$ between measurements was assumed based on reported correlations in Hacker et al. (1992); McNamara et al. (2011), and Sánchez et al. (2017).⁶

3.3. Analysis

We compared four multilevel meta-analytic models (see Fig. 1) through a stepwise likelihood ratio test. Model 1 was a standard random-effects model with sampling variance (extracted from the studies) at Level 1 and the random variance of effect sizes at Level 2. For Model 2, we added a third level to account for the nesting of effect sizes within studies. Model 3 included an intermediate level that accounted for the nesting of effect sizes within subgroups, which were in turn nested within studies. Model 4 was a cross-classified random-effects model (CCREM, Fernández-Castilla et al., 2019), with assessment type as the cross-level random effect in addition to the hierarchical Levels 2 to 4.

For the meta-regression model including linguistic modifications, the intercept was fixed at zero; because all studies were experimental studies, the hypothetical absence of any modification would necessarily lead to a null effect. Thus, in cases where studies included several linguistic modifications, effects of these modifications add up to the total estimated effect.

All models were fitted using the R package *metafor* (Viechtbauer, 2010). Confidence intervals of all estimates were calculated based on a t -distribution, similar to the Knapp and Hartung adjustment (Knapp & Hartung, 2003; Viechtbauer, 2010). I^2 (representing the proportion of variation due to heterogeneity) was calculated according to the method described in Viechtbauer (2022; see also Cheung, 2014; Higgins & Thompson, 2002). R^2 values were calculated as described in Cheung (2014). Tests for residual heterogeneity and omnibus tests of moderators were based on F-distributions (Viechtbauer, 2010).

4. Results

4.1. Descriptive results

Our analysis included 45 studies, consisting of 79 subsamples with a total of $N = 6477$ learners. We extracted 188 effect sizes from these studies. An overview of all studies, the most relevant moderators and the corresponding effect sizes is given in Table A.3.

4.2. Overall effect size

The model fit comparison of the four random effects models is given in Table 2. The results show that Model 3 with a hierarchical four-level structure fitted the data best. For this model, the mean effect of linguistic modifications was $g = 0.147$, $SE = 0.035$, $p < .001$, 95% CI [0.077, 0.216]. The prediction interval, [-0.43, 0.72], and a Q-test, $Q(187) = 522.85$, $p < .0001$ indicated substantial heterogeneity between the studies. I^2 was moderate for Level 2 ($I^2_{(2)} = 21.3\%$) and Level 3 ($I^2_{(3)} = 44.3\%$), no variance was explained by heterogeneity on the between-studies-level ($I^2_{(4)} = 0.0\%$).

4.3. Meta-regression

For each set of moderators, separate meta-regression analyses were conducted using the hierarchical four-level random-effects structure from Model 3 (as described in Section 6.2), with moderators added with fixed effects. Results are given in Table 3.

4.3.1. Prior knowledge

Prior knowledge was a significant moderator of effect sizes, $F(2,185) = 9.122$, $p < .001$. Together, content prior knowledge and language prior knowledge explained 55% of the variance at the between-subgroup level, but no variance on the other levels. Only content prior knowledge predicted the effect size significantly, $b = -0.182$, $p < .001$.

4.3.2. Linguistic modifications

The eight categories of linguistic modifications significantly moderated effect sizes, $F(8,180) = 3.777$, $p < .001$. These moderators explained 10% of the variance on the between-effects level and 74% of the variance on the between-studies level. However, of the eight categories, only *increasing local clarity and elaboration*, *increasing global clarity and elaboration/summaries*, and *increasing global clarity and elaboration/personalization* significantly predicted effect sizes, all with positive effects, $b = 0.200$, $b = 0.146$, and $b = 0.522$, $p = .009$, $p = .011$, $p = .002$. When interaction terms with prior knowledge were included, positive effects of these categories were still significant, with significant interaction effects between prior knowledge and categories *increasing global clarity and elaboration/summaries*, and *increasing global clarity and elaboration/personalization*, such that these modifications were more effective for participants with low prior knowledge. The inclusion of the eight interaction terms had an overall marginally significant influence, $F(8,172) = 1.995$, $p = .0497$.

The interaction model for linguistic modifications with language prior knowledge indicated that none of the interaction effects were significant individually ($p > .30$) or overall $F(7,173) = 0.375$, $p = .916$.

⁶ Since these tasks are assumed to measure the same outcome, their correlation can be interpreted as reliability coefficient. Accordingly, the value of $r = 0.80$ also coincides with what is generally considered the threshold for a good reliability.

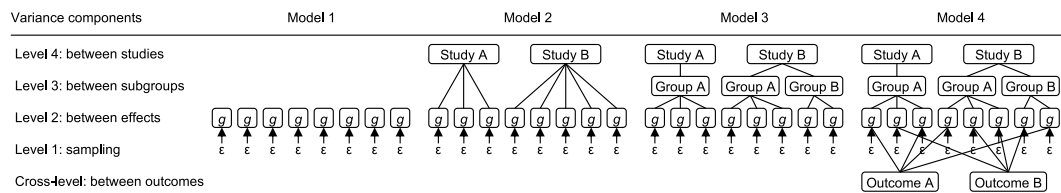


Fig. 1. Structure of the tested models.

Table 2

Likelihood ratio tests for the four tested models.

	Variance (τ^2)				Estimate (SE)	AIC	BIC	LRT
	L2	L3	L4	cross-level				
Model 1	.091				0.13 (0.028)	201.26	207.72	
Model 2	.055		.030		0.17 (0.038)	186.94	196.63	16.32***
Model 3	.027	.056	.000		0.15 (0.035)	173.95	186.88	14.98***
Model 4	.027	.056	.000	.000	0.15 (0.035)	175.95	192.11	0.00

Note. L2 = between effects, L3 = between subgroups, L4 = between studies, cross-level = between outcomes. AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, LRT = Likelihood ratio test. The LRT refers to the previous model, respectively.

*** $p < .001$.

4.3.3. Outcome measure

The comparison between recall, apply, and transfer measures did not indicate a significant influence of the outcome measure on effect sizes, $F(2,185) = 0.086$, $p = .918$.

4.4. Study descriptors

The year of publication was not a significant moderator of effect size, $F(1,168) = 1.350$, $p = .247$. Although the omnibus test of text domain as a moderator was not significant, $F(4,183) = 1.625$, $p = .170$, studies in the domain of physics showed significantly higher effects than studies with a mixed science context as a reference category, $b = 0.288$, $p = .016$.

Text length was a significant moderator of effect size, $F(1,186) = 6.040$, $p = .015$. For each 100 words, the estimated effect size increased by $g = 0.017$. The text language, age group or study location did not significantly moderate effect size, $p > .366$.

4.5. Publication bias

According to [Rodgers and Pustejovsky \(2020\)](#), existing standard procedures to check for publication bias (including analyses of funnel plot asymmetry, trim-and-fill analyses, or Egger's regression, see [Borenstein et al., 2009](#)) do not give reliable results for multilevel meta-analyses. An alternative approach is to compare the mean effect size of non-reviewed *grey literature* against peer-reviewed publications ([Borenstein et al., 2009](#)). In our dataset, only two studies were not published in peer reviewed papers, so a systematic comparison by this criterion was not meaningful. Instead, we isolated publications that were either grey literature or did not report a main effect of a linguistic modification in the abstract. Our rationale was that studies that did not focus on the linguistic modification would, similar to grey literature, not be subject to a systematic selection based on the effect size of this modification. Of the 45 studies, 15 did not refer to a main effect of linguistic modification in their abstract or had not been subject to peer-review. The mean effect size in these studies ($g = 0.11$) did not differ significantly from the mean effect size reported in the peer-reviewed studies ($g = 0.16$), $F(1,186) = 0.411$, $p = .522$. Thus, we did not find evidence for a publication bias.

5. Discussion

In this meta-analysis, we summarized empirical research findings on the effect of linguistic modifications of STEM texts on comprehension and learning. Regarding our first research question, we found a small overall effect of $g = 0.15$. Even though linguistically modified texts led to better comprehension and learning in most cases, the size of the average effect indicates that this improvement was not particularly strong compared to other approaches, such as providing feedback ($g = 0.35$; [Swart et al., 2019](#)) or including graphics ($g = 0.39$; [Guo et al., 2020](#)). Linguistic modifications in STEM texts are relevant, but not a guarantee for strong improvements in comprehension and learning.

5.1. Moderators of the effect of linguistic modifications

5.1.1. Content prior knowledge

Our second research question focused on how prior knowledge affects the outcome of linguistic modifications in STEM texts.

Table 3

Results of meta-regressions for all moderators.

Effect	Estimate	SE	95% CI		p	I ²	R ²
			LL	UL			
No fixed effects							
Intercept	0.147	0.035	0.077	0.216	<.001		
Random effects							
Between-effect variance	0.027		0.009	0.054		21.33	
Between-subgroup variance	0.056		0.023	0.094		44.26	
Between-study variance	0.000		0.000	0.029		0.00	
Prior knowledge							
Fixed effects							
Intercept	0.170	0.036	0.098	0.242	<.001		
Content prior knowledge ^a	−0.182	0.043	−0.266	−0.097	<.001		
Language prior knowledge ^a	0.031	0.056	−0.079	0.141	.577		
Omnibus-test F(2,185)	9.122				<.001		
Random effects							
Between-effect variance	0.030		0.011	0.058		26.27	0.00
Between-subgroup variance	0.025		0.000	0.069		21.84	55.06
Between-study variance	0.017		0.000	0.051		14.51	0.00
Linguistic modifications							
Fixed effects							
Red. word level complexity	0.095	0.122	−0.146	0.336	.438		
Red. syntactic complexity	−0.018	0.118	−0.250	0.214	.881		
Incr. local cohesion							
Overlap	−0.076	0.080	−0.233	0.082	.343		
Connectives	−0.071	0.064	−0.196	0.055	.269		
Incr. global cohesion	−0.016	0.097	−0.207	0.175	.871		
Incr. local clarity and elaboration	0.200	0.076	0.050	0.350	.009		
Incr. global clarity and elaboration							
Summaries	0.146	0.057	0.034	0.258	.011		
Personalization	0.522	0.167	0.194	0.851	.002		
Omnibus-test F(8,180)	3.777				<.001		
Random effects							
Between-effect variance	0.024		0.006	0.051		19.21	9.77
Between-subgroup variance	0.058		0.029	0.098		45.68	0.00
Between-study variance	0.000		0.000	0.025		0.00	74.27
Interaction between linguistic modifications and content prior knowledge							
Fixed effects							
Red. word level complexity	0.080	0.125	−0.167	0.327	.525		
Red. syntactic complexity	0.003	0.119	−0.232	0.238	.981		
Incr. local cohesion							
Overlap	−0.072	0.081	−0.232	0.087	.373		
Connectives	−0.069	0.064	−0.194	0.057	.281		
Incr. global cohesion	−0.016	0.097	−0.207	0.174	.865		
Incr. local clarity and elaboration	0.195	0.076	0.045	0.345	.011		
Incr. global clarity and elaboration							
Summaries	0.155	0.057	0.042	0.268	.007		
Personalization	0.377	0.172	0.037	0.717	.030		
Fixed effects: interactions ^a							
Red. word level complexity	−0.278	0.434	−1.135	0.579	.523		
Red. syntactic complexity	0.392	0.515	−0.624	1.407	.448		
Incr. local cohesion							
Overlap	−0.022	0.175	−0.367	0.323	.898		
Connectives	−0.024	0.156	−0.332	0.283	.876		
Incr. global cohesion	0.141	0.128	−0.111	0.393	.271		
Incr. local clarity and elaboration	−0.100	0.135	−0.367	0.167	.459		
Incr. global clarity and elaboration							
Summaries	−0.126	0.062	−0.249	−0.003	.045		
Personalization	−0.521	0.209	−0.934	−0.109	.014		
Wald-type χ^2 test F(8,172) ^d	1.995				.050		
Omnibus-test F(16,172)	2.952				<.001		
Random effects							
Between-effect variance	0.025		0.007	0.053		20.84	5.61
Between-subgroup variance	0.052		0.017	0.092		42.41	7.45
Between-study variance	0.000		0.000	0.034		0.00	0.00

(continued on next page)

Table 3 (continued)

Effect	Estimate	SE	95% CI		p	I ²	R ²
			LL	UL			
Interaction between linguistic modifications and language prior knowledge							
Fixed effects							
Red. word level complexity	0.076	0.128	−0.176	0.329	0.552		
Red. syntactic complexity	−0.002	0.122	−0.242	0.238	0.987		
Incr. local cohesion							
Overlap	−0.077	0.081	−0.237	0.084	0.348		
Connectives	−0.073	0.065	−0.202	0.055	0.262		
Incr. global cohesion	−0.017	0.100	−0.214	0.180	0.862		
Incr. local clarity and elaboration	0.205	0.077	0.052	0.357	0.009		
Incr. global clarity and elaboration							
Summaries	0.143	0.061	0.023	0.263	0.019		
Personalization	0.523	0.170	0.188	0.858	0.002		
Fixed effects: interactions ^d							
Red. word level complexity	0.228	0.277	−0.319	0.775	0.412		
Red. syntactic complexity	−0.232	0.291	−0.806	0.342	0.427		
Incr. local cohesion							
Overlap	0.192	0.274	−0.349	0.734	0.484		
Connectives	0.116	0.124	−0.128	0.360	0.351		
Incr. global cohesion	−0.063	0.189	−0.435	0.310	0.740		
Incr. local clarity and elaboration	−0.266	0.258	−0.776	0.244	0.304		
Incr. global clarity and elaboration							
Summaries	0.038	0.164	−0.286	0.361	0.819		
Personalization	no variation of language prior knowledge						
Wald-type χ^2 test F(7,173) ^d	0.375				.916		
Omnibus-test F(15,173)	2.121				.011		
Random effects							
Between-effect variance	0.026		0.007	0.054		19.57	3.81
Between-subgroup variance	0.061		0.031	0.106		46.43	0.00
Between-study variance	0.000		0.000	0.025		0.00	76.30
Assessment outcome							
Fixed effects							
Intercept (RC: Recall)	0.145	0.045	0.056	0.233	.001		
Apply ^b	0.010	0.053	−0.094	0.114	.849		
Transfer ^b	−0.022	0.077	−0.173	0.129	.773		
Omnibus-test F(2,185)	0.086				.918		
Random effects							
Between-effect variance	0.028		0.009	0.056		21.77	0.00
Between-subgroup variance	0.056		0.023	0.095		44.13	0.00
Between-study variance	0.000		0.000	0.031		0.00	0.00
Publication year							
Fixed effects							
Intercept (2000)	0.131	0.037	0.056	0.205	<.001		
Year	0.004	0.003	−0.003	0.001	.247		
Omnibus-test F(1,186)	1.350				.247		
Random effects							
Between-effect variance	0.027		0.009	0.054		21.60	0.00
Between-subgroup variance	0.055		0.025	0.093		43.74	1.41
Between-study variance	0.000		0.000	0.027		0.00	32.88
Domain							
Fixed effects							
Intercept (RC: mixed science)	0.058	0.061	−0.063	0.179	.342		
Physics	0.288	0.119	0.054	0.522	.016		
Biology	0.113	0.081	−0.047	0.274	.166		
Computer sciences	0.015	0.137	−0.254	0.285	.911		
Other	0.085	0.153	−0.217	0.387	.581		
Omnibus-test F(4,183)	1.625				.170		
Random effects							
Between-effect variance	0.028		0.009	0.056		22.73	0.00
Between-subgroup variance	0.052		0.022	0.091		42.00	6.60
Between-study variance	0.000		0.000	0.026		0.00	56.47
Text length							
Fixed effects							
Intercept	0.038	0.056	−0.072	0.148	.496		

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Table 3 (continued)

Effect	Estimate	SE	95% CI		p	I ²	R ²
			LL	UL			
Text length ^c	0.017	0.007	0.003	0.030	.015		
Omnibus-test $F(1,186)$	6.040				.015		
Random effects							
Between-effect variance	0.027		0.009	0.054		21.94	0.45
Between-subgroup variance	0.052		0.024	0.089		42.46	7.16
Between-study variance	0.000		0.000	0.023		0.00	70.78
Text language							
Fixed effects							
Intercept (RC: Other languages)	0.202	0.066	0.071	0.333	.003		
English	−0.096	0.080	−0.254	0.063	.235		
German	0.021	0.125	−0.225	0.267	.865		
Omnibus-test $F(2,185)$	1.010				.366		
Random effects							
Between-effect variance	0.028		0.009	0.055		21.99	0.00
Between-subgroup variance	0.055		0.024	0.093		43.30	2.40
Between-study variance	0.000		0.000	0.028		0.00	50.34
Age group							
Fixed effects							
Intercept (RC: Post-secondary)	0.138	0.048	0.044	0.231	.004		
Primary	−0.036	0.114	−0.262	0.189	.753		
Lower Secondary	0.001	0.088	−0.164	0.183	.913		
Upper Secondary	0.136	0.134	−0.128	0.400	.310		
Omnibus-test $F(3,184)$	0.419				.739		
Random effects							
Between-effect variance	0.027		0.009	0.054		21.06	0.00
Between-subgroup variance	0.058		0.023	0.098		45.09	0.00
Between-study variance	0.000		0.000	0.034		0.00	0.00
Study Location							
Fixed effects							
Intercept (RC: Laboratory)	0.102	0.047	0.010	0.195	0.030		
Classroom, irregular setting	0.119	0.112	−0.102	0.339	0.290		
Regular classroom setting	0.083	0.083	−0.080	0.246	0.317		
Online	0.181	0.178	−0.170	0.532	0.311		
Omnibus-test $F(3,184)$	0.795				0.498		
Random effects							
Between-effect variance	0.027		0.009	0.054		21.27	0.00
Between-subgroup variance	0.056		0.022	0.095		43.92	0.00
Between-study variance	0.000		0.000	0.032		0.00	0.00

Note. Number of studies = 45, number of subgroups = 79, number of effects = 188, total $N = 6477$. CI = confidence interval; LL = lower limit; UL = upper limit, RC = reference category.

^a -1 = low, 0 = regular, 1 = high.

^b 0 = no, 1 = yes. For 16 (8.5%) effects referring to combinations of more than one outcome, each outcome was coded 1/2 = yes (two outcomes) or 1/3 = yes (three outcomes), respectively.

^c per 100 words.

^d The Wald-type test refers to the eight interaction effects in total.

Previous research had identified a form of an *expertise reversal effect* (Kalyuga, 2003) for the specific case of increasing text cohesion, which is called the *reverse cohesion effect* (e.g., McNamara et al., 1996). In the case of linguistic modifications, the expertise reversal effect refers to the idea that readers with low content prior knowledge benefit from linguistic modifications, while the modifications could lose their effectiveness or even have negative effects for readers with high content prior knowledge. The content of STEM texts can be very demanding (Alexander et al., 1994; Best et al., 2005; Follmer & Sperling, 2018), and there can be large differences in prior knowledge among readers (Graesser et al., 2002), which could mean that the effect is particularly strong for STEM texts.

Our meta-analysis allowed us to quantify this effect across all included studies and all linguistic modifications: For low prior-knowledge readers, the mean effect size of linguistic modifications was substantial ($g = 0.35$), whereas for high prior knowledge groups, it was almost zero ($g = -0.01$). This strongly supports the existence of an expertise reversal effect. However, it should be emphasized that linguistic modifications did not have a relevant *negative* effect for any readers.

There are several possible explanations for this effect. McNamara et al. (1996) suggested that high prior knowledge students process texts only superficially when they do not offer them challenging gaps in cohesion, which could negatively impact comprehension and learning. According to Cognitive Load Theory, lower prior knowledge might lead to higher intrinsic load (Sweller et al., 2019; van Merriënboer & Sweller, 2005), and limit germane load, which is something that linguistic modifications aiming at reducing extraneous load (e.g., by decreasing linguistic complexity) can possibly compensate for. In contrast, high prior knowledge readers

might not experience an overload in the first place, and thus, might not benefit from reducing extraneous load. Inducing germane load (e.g., by making the text more explicit or adding elaborations) might be redundant or even have negative effect by misdirecting attention or decreasing engagement (Kalyuga et al., 2003; McNamara et al., 1996). Future studies could further consider the role of cognitive load during reading of STEM texts, providing a clearer explanation which processes during comprehension and learning from texts are affected by linguistic modifications, and their relation to prior knowledge.

5.1.2. Language prior knowledge

Our meta-analysis did not show an interaction of language prior knowledge with the effect of linguistic modifications. Possibly, this interaction is more complex and depends on other factors which were not considered in our analyses. For example, Ozuru et al. (2009) and O'Reilly and McNamara (2007) suggested a three-way interaction between language prior knowledge, content prior knowledge, and assessment type. Another explanation might be that the groups compared here did not differ enough regarding their language prior knowledge. Linguistic modifications might be particularly beneficial for readers with very low language prior knowledge which struggle with basic reading processes, but similarly effective for readers above a certain level. In the studies included in this analysis, readers with low language prior knowledge were usually identified through percentiles or median splits, but the overall level of these groups was not specified. Only two studies used norms of standardized test scores to identify low language prior knowledge (Geva & Ryan, 1985; Meyer et al., 1980), and only one study (Degand & Sanders, 2002) compared first and second language learners. Therefore, it is possible that in the studies included here, the language prior knowledge of the participants was not low enough to interact with linguistic modifications, that relevant subgroups were not identified by the applied procedures, or that our analytical design did not account for relevant interactions.

5.1.3. Categories of linguistic modifications

Regarding research question three, we studied the individual impact of eight categories of linguistic modifications on comprehension and learning. Previous studies often modified multiple linguistic features simultaneously, making it difficult to compare results (Cruz Neri & Retelsdorf, 2022). Our analysis included all categories in one meta-regression model, which allowed us to isolate their average effects.

Only three categories were found to have a significant impact on comprehension and learning: Increasing clarity and elaboration on the word and sentence level, increasing clarity and elaboration on the global text level, and personalization. All three categories aimed to help readers make inferences and connections to prior knowledge and elaborate on key concepts (see Sect. 2).

Both increasing clarity and elaboration locally and globally showed relatively small effects on comprehension and learning, but for studies combining these modifications the influence adds up to a moderate-sized effect ($g = 0.35$). In contrast to the other modifications, studies that increased clarity and elaboration took into account the content of the text and typically added content information (McNamara et al., 2010). In some studies, this increased text length by up to 50% (e.g., McNamara et al., 1996; McNamara et al., 2011; Ozuru et al., 2009; Vidal-Abarca & Sanjose, 1998) or even 100% (Gasparinatou & Grigoriadou, 2011). Thus, it is important to critically discuss the line between linguistic modification and content modification. Our categorization considered the modifications to be linguistic in nature, since the amount of information regarding the target content (which should be comprehended and learned and which was assessed afterwards) remained constant while the surrounding text was changed to foster comprehension and learning of this content. However, the particular effectiveness of these modifications might indicate that modifications on the linguistic level are generally more helpful when they are interlaced with improvements on the content level. It might be particularly successful to use linguistic modifications for structuring and focusing the reading processes towards the content-specific learning goals.

On the local level, this involves clarifying and elaborating important aspects of the content. On the global level, content-specific considerations should determine the structure, headings, and content of summaries to maximize comprehension and minimize distractions from irrelevant or redundant information, as shown in research on seductive details (Sundararajan & Adesope, 2020) and the redundancy effect in multimedia learning (Mayer et al., 2001).

Personalization of texts had the strongest positive effect in our analyses. This result was based on three studies (Dutke et al., 2016; Moreno & Mayer, 2000; Stiller & Jedlicka, 2010) which focused only on personalization. This implies a very reliable effect which is clearly attributable to this particular modification. Personalization seems to offer a promising potential for increasing text comprehension compared to other linguistic modifications.

Somewhat surprisingly, the remaining five categories of linguistic modifications did not have a significant effect on comprehension and learning. This finding may be specific to STEM texts, where deeper conceptual understanding is particularly important compared to narrative texts (van den Broek, 2010). In general, the nature of the content of STEM texts might limit the possibilities of linguistic modifications without making concessions to the content. Often, the complexity of the language mirrors the complexity of the content and cannot easily be simplified or changed (Schleppegrell, 2001). For example, the use of technical vocabulary provides difficulties for many readers, but using everyday language is usually not viable for scientific precision. The repeated use of the same words instead of synonyms, which is often considered to increase cohesion and support text comprehension, might be counterproductive when technical terminology is introduced (van den Broek, 2010). Accordingly, some of the approaches to make other texts easier to comprehend might simply not be transferable to STEM texts, or might not address the aspects of the text which are relevant for content-specific comprehension and learning.

In addition, Cognitive Load Theory also offers some explanations for the varying effects of the linguistic modifications. It suggests that extraneous load can be reduced by decreasing the cognitive resources that are only required for reading, but not for the processing of the STEM content. For example, increasing cohesion is assumed to reduce the amount of information that has to be kept in working memory in order to infer connections between sentences (van den Broek, 2010). The fact that these linguistic modifications were not

effective could indicate that, for the readers included here, the reading process did not cause enough extraneous load to meaningfully interfere with intrinsic and germane load. In contrast, the effective linguistic modifications that aim at increasing links with prior knowledge did not aim at reducing extraneous load, but at directing germane load effectively towards inferences with prior knowledge (see also [Leutner et al., 2009](#); [Sweller et al., 2019](#); [van Merriënboer & Sweller, 2005](#)). Possibly for reading STEM texts under regular circumstances and by proficient readers, this could be more effective than freeing up resources by reducing extraneous load.

Finally, these analyses are based on a simplified, additive model. Arguably, linguistic features interact with each other, causing more complex interrelations which are not captured by our model. Therefore, future studies might vary different linguistic text features more systematically, and investigate interactions between them.

5.1.4. Extent of linguistic manipulations

In the current meta-analyses, we summarized the effects of original texts versus modified texts without taking the extent of the linguistic modifications into account. The main challenge in quantifying linguistic manipulations is the lack of a common metric. Many studies did not report a metric to quantify their modifications, and if they did, the metrics differed across the categories of linguistic modifications.

For example, studies that aimed to increase cohesion often referred to the *Coh-Metrix* ([Graesser et al., 2011](#)), a tool that provides a variety of metrics of text characteristics. Some of these studies reported up to 8 detailed characteristic metrics of their texts (e.g., [McNamara et al., 2011](#); [Ozuru et al., 2009](#); [Ozuru et al., 2010](#)). In comparison, studies that modified the use of connectives or simplified words usually reported the absolute number of changes to quantify the modification, or gave specific metrics like the average lexical frequency of the modified words ([Arya et al., 2011](#)). Arguably, there is no scientifically reasonable way to compare these values with scores from the *Coh-Metrix*, which would be necessary in order to include the extent of modifications in a meta-regression as an additional continuous predictor across all modifications. For other modifications like making the top-level structure explicit before the text, it is hard to come up with a way to quantify the changes at all.

One common metric, the *Flesch-Kincaid Grade Level* (FKGL; [Flesch, 1948](#); [Kincaid et al., 1975](#)) was even used for different purposes in different studies. The FKGL takes into account the number of words, sentences, and syllables in a text. In [Burton \(2014\)](#), the linguistic modification aims at increasing readability; therefore, the FKGL was used as a metric to quantify this modification (FKGL (original) ≈ 11.5 ; FKGL(modified) ≈ 8.9). [Follmer and Sperling \(2018\)](#) aimed at increasing referential cohesion, using FKGL as a control metric to ensure that the modifications did not affect readability (FKGL = 8 in both conditions, see also [Lippmann et al., 2019](#)). In contrast, increasing cohesion will generally increase the number of words per sentence, which in turn leads to lower readability metrics ([McNamara, 2001](#)). Accordingly, studies on cohesion typically reported lower readability metrics after increasing cohesion, for example in [O'Reilly and McNamara \(2007\)](#); FKGL(low cohesion) = 9.2; FKGL(high cohesion) = 10.5; see also [McNamara et al., 2011](#); [Ozuru et al., 2010](#); [Ozuru et al., 2009](#)

In conclusion, meaningfully comparing the extent of linguistic modifications across different categories seems to be hardly possible. Therefore, the reported effects should be understood as averages, and it is possible that some of the modifications could differ in their effect depending on the extent of their implementation.

5.1.5. Expertise reversal effect for specific linguistic modifications

We further investigated whether the expertise reversal effect for content prior knowledge differed between the eight categories of linguistic modifications. However, with a total of 16 predictors and the substantial heterogeneity across studies, this analysis should not be overinterpreted. A careful interpretation points towards the finding that the expertise reversal effect for content prior knowledge might differ between linguistic modifications: While increasing personalization and global clarity was significantly more effective for low prior-knowledge groups of participants, no evidence towards the effect was found in the other categories. For language prior knowledge, no significant interactions were found. Thus, while the possibility of an expertise reversal effect should always be considered when modifying STEM texts, it might not occur under certain circumstances (see also [Follmer & Sperling, 2018](#)).

5.1.6. Study characteristics

Regarding research question four, we investigated which other study characteristics influenced the effects of linguistic modifications.

Our analysis showed no indication that linguistic modifications were more effective for measures of recall, application, or transfer, even though these three measures address different aspects of text comprehension ([McNamara & Magliano, 2009](#)). It was notable that many studies primarily assessed measures of recall which often asked for a verbatim reproduction of superficial information from the text. In line with [van den Broek \(2010\)](#), we found that relatively little research focused on permanent learning.

Modifications of physics texts had stronger effects on comprehension and learning than in other domains. However, these texts covered a wide range of topics, contexts, and age groups which makes it difficult to identify a clear explanation. Possibly, the texts that were explicitly situated in physics might have been more demanding. A higher content demand would be equivalent to lower prior knowledge, which could support the success of linguistic modifications.

The longer the texts in the summarized studies, the stronger the effect of linguistic modifications were. When very short texts are compared with text of multiple pages, this adds up to a substantial difference, with an increase in effect size of $g = 0.017$ per 100 words or about $g = 0.076$ per page. Possibly, longer text put higher demands on cognitive resources, required a more extensive mental representation, and included more distant semantic relations.

5.2. Implications

Texts play an important role in STEM education, and the selection, design and revision of texts is a common practice for educators. Thus, we believe that some consequences for educational practice and research can be drawn from the analyses reported in this study.

The results of the study showed a positive impact from linguistic modifications, but the magnitude of this effect was relatively small compared to the well-established benefits of other text modifications such as providing feedback or incorporating graphics (Guo et al., 2020; Swart et al., 2019). The effectiveness of these modifications was also dependent on the reader's prior knowledge. Therefore, relying solely on linguistic modifications may not lead to significantly improved learning outcomes for all students. In turn, if practitioners do not have the knowledge or tools for improving texts linguistically (as might be the case for some STEM educators), this will probably not have devastating effects on learning from one particular text. However, if these challenges are ignored over a longer period of time, the effects could accumulate.

Our analysis showed that readers with low content-specific prior knowledge could be supported very effectively by linguistic modifications. Thus, it is important for practitioners to remember that linguistic modifications not only support reading, but also learning and comprehension of the content. In practical terms, teachers should consider making linguistic modifications not only when students face challenges with language, but also if they are having difficulties with the content.

Our study found that two of the most effective modifications were related to the content and were implemented using linguistic tools such as clarifying the content and adding elaborations. Theoretically, this helps integrate the text with readers' prior knowledge and aids in building a meaningful understanding of the material (McNamara & Magliano, 2009). Practically speaking, this suggests that language modifications should aim not just to improve the surface-level features of the text, but to enhance its overall comprehensibility of the content. For example, only using simpler words or shorter sentences may not be as helpful as adding additional information to elaborate on technical terms and key concepts or including summaries at the end of paragraphs. In other words, teachers should not focus solely on linguistic features, but on how to use domain-specific language effectively to support comprehension of the content (Schleppegrell, 2001).

It's important to note that comprehension and learning might not be the only objectives when working with STEM texts. The design of a text will likely involve a consideration of other factors that are not discussed in this analysis. For example, texts for younger students might serve to enhance reading comprehension skills, where a degree of challenge is desired. The use of linguistically complex texts could also prepare students for future challenges in STEM studies or careers, as it trains cognitive and metacognitive strategies for handling difficult texts. Authentic, technical language in texts might also be motivating for some students when it reflects the real-world science they are interested in, but it might be frustrating for others.

5.3. Conclusion

Texts play an important role for learning in STEM domains, and various researchers and practitioners have sought to identify linguistic features that influence learning from these texts. Based on this meta-analysis, we conclude that linguistic modifications can be a tool to improve texts for learning, but only to a limited extent and under certain circumstances. It seems that modifications on the surface level have, in general, rather small effects. In contrast, integrating and coordinating efforts to improve the language and the content of texts simultaneously can support learning effectively, in particular in longer texts and for learners with low content prior knowledge.

Author statement

Anselm R. Strohmaier: Methodology, Formal Analysis, Data Curation, Writing – Original Draft. Timo Ehmke: Conceptualization, Methodology, Data Curation, Writing – Review & Editing. Hendrik Härtig: Conceptualization, Data Curation, Writing – Review & Editing. Dominik Leiss: Conceptualization, Data Curation, Writing – Review & Editing.

Declarations of competing interest

None.

Table A.1

search string for database queries

	AND				
	Text characteristics	Outcome	STEM	Experimental design	Material
OR	language demand	achievement	mathematics	experiment*	text*
	readability	competence	science	intervention*	word problem*
	text comprehension	performance	physics	control group	test item*
	language difficulty	assessment*	biology	laboratory study	
	linguistic complexity	understand*	chemistry	influenc*	
	coherence	comprehen*	scientific	interaction*	

(continued on next page)

Table A.1 (continued)

AND					
	Text characteristics	Outcome	STEM	Experimental design	Material
	syntactic complexity	acquisition	science-related	connection*	
	lexical complexity	learn*		impact*	
	high-coherence			effect*	
	low-coherence			expt*	
	text structure			groups	
	refutation*			association	
	cohesive				
	functional linguistics				
	cohesion				
	language demands				

Table A.2
Condensed coding scheme for categories of linguistic modifications

Category	Description
1 Reducing word-level complexity	Reducing complexity at the word level is achieved by using more frequent or shorter words and avoiding technical terms. In addition, complex words such as compounds, appositions, attributes, modifiers, nominalizations and complex adjectives are avoided. Personalization (6b) does not belong in this category, as it is not primarily aimed at simplification at the word level. Changes that alter text cohesion (connectors, pronouns) also do not belong in this category (but 3a or 3b). In this category, only individual words or phrases are changed or omitted or their frequency is changed; additional explanations or additions are Category 5. Examples: "replacing technical terms with less specialized vocabulary" "simpler words" Counterexamples: "replacing nouns with pronouns" (3a) "use second person pronouns" (6b)
2 Reducing syntactic complexity	Syntactic complexity is reduced by avoiding complex and long sentence structures, for example by using short main clauses instead of complex and long subordinate clause structures. The use of passive voice is also avoided. The category also includes the avoidance of a complex sentence structure (not text structure) through word order, for example when deviating from the prototypical subject-predicate-object structure. This category does not include additions to content such as headings, introductions, etc. (6a). The linking of sentences by connectors is also not rated as a change in syntactic complexity (3b). Examples: "reducing relative clauses, simplify syntax" "using shorter sentences" Counterexamples: "rearrange sentences" (4) "link sentences through connective phrases" (3b)
3a Increasing local cohesion/overlap	Local cohesion tries to create an overlap for local sentence elements (at the word level) by linguistic means. This is achieved by using nouns instead of pronouns and by repeating words instead of using synonyms (not as repetition in the sense of repeating content, which is Category 5). Examples: "use nouns instead of pronouns" "use same words instead of synonyms". Counterexamples: "use of possessive pronouns" (6b) "use of connectives" (3b)
3b Increasing local cohesion/connectives	This category links sentence elements through the use of connectors or connective phrases. This includes the insertion, marking or systematic omission of various connectors. Connective phrases ("For this reason ...") also fall into this category. Examples: "connective phrases" "making connections between sentences explicit" Counterexamples: "increase overlap" (3a) "add linking macropositions" (6a)
4 Increasing global cohesion	This category links sentence elements through structural changes. This happens exclusively by changing the structure and arrangement of the same content, but no additions are made to the content (which is 5 or 6a). This includes changing the order of sentences or paragraphs, for example into a chronological or factual order. Examples: "change sentence order" "rearrange sentences" Counterexamples:

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Table A.2 (continued)

Category		Description
5	Increasing local clarity and elaboration	"advanced organizers" (6a)
		"add elaborations to make the structure clearer" (6a)
		Local clarity and elaboration are achieved through content additions and content repetitions at the word and sentence level. For this purpose, inferences are explained, relations and actions are made explicit. This also includes the insertion of relative pronouns (this makes previously not apparent connections explicit, i.e., adds information), the insertion of explanations (also when they have a cohesive function "adding connective elaborations"), and the explicit linking to prior knowledge. In this category, information is explicitly added or repeated; in Categories 3a, 3b, and 4 unchanged content is only linked differently. This category does not include the replacement of difficult technical terms with simple words (1) or the explication of inferences through connectors or the avoidance of pronouns (3a, 3b). Similarly, merely repeating single words across sentences is Category 3a. Examples: "repetition of information" "replacing ambiguity" Counterexamples: "repeat noun instead of pronoun" (3a) "more overlapping words" (3a)
6a	Increasing global clarity and elaboration/summaries	Global clarity is achieved through content additions or clarity to the global text structure and content. These additions can be on a mainly structural level in terms of content (e.g., headings) or make substantive additions to the text content (e.g., summaries). This category includes headings, subheadings, titles, topic sentences, previews and reviews, summaries, macropropositions, advanced organizers and objective clarifications. Like category 5, category 6a is distinguished from categories 3a, 3b, and 4 by the fact that in 6a, information is added (or repeated). This category does not include elaborations on individual content elements and words (often referred to as 'adding elaborations', which is 5). Examples: "summaries added" "adding topic headers" Counterexamples: "paragraphs reorganized" (4) "change order of sentences" (4)
6b	Increasing local clarity and elaboration/personalization	This category includes personalization through the use of second person personal pronouns and possessive pronouns instead of definite articles. Example: "possessive pronouns"

Table A.3

Study characteristics and effect sizes of all included studies

Authors	Year	Group	N	Domain	Text length	Age group	Prior Knowledge		Outcome			Linguistic Modification						g	Var		
							Content	Language	Recall	Apply	Transfer	1	2	3a	3b	4	5			6a	6b
Arya, Hiebert, & Pearson	2011	1	142	MIX	200	pri	0	0	X	X		X	X						0.12	0.024	
							0	0	X	X										0.15	0.024
Boscolo & Mason	2003	1	69	PHY	1390	sec2	–	0	X					X	X				–0.31	0.080	
							–	0		X				X	X				–0.12	0.072	
							–	0			X			X	X				–0.07	0.079	
							–	0	X								X		0.64	0.095	
							–	0		X							X		1.00	0.092	
							–	0			X						X		0.88	0.099	
							+	0	X					X	X				–0.28	0.064	
							+	0		X				X	X				0.00	0.057	
							+	0			X			X	X				0.03	0.063	
							+	0	X								X		0.15	0.065	
Burton	2014	1	56	BIO	200	p-sec	+	0		X								X		0.32	0.059
							+	0			X							X		0.09	0.065
							0	+	X	X	X	X	X						0.25	0.098	
							0	+	X	X	X	X	X						0.12	0.103	
							0	0		X						X			0.49	0.029	
							0	0		X					X				0.35	0.027	
							0	0	X							X			0.56	0.029	
							0	–	X						X				0.36	0.027	
							0	–	X						X				0.42	0.027	
							0	–	X						X				0.31	0.026	
Deppert	2001	1	21	BIO	400	p-sec	–	0		X	X		X	X	X	X		0.80	0.083		
							+	0		X	X		X	X	X	X		–0.76	0.089		
Dutke, Grefe, & Leopold	2016	1	65	BIO	915	sec2	0	0		X							X	0.03	0.054		
							0	0			X							X	1.11	0.063	

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Table A.3 (continued)

Authors	Year	Group	N	Domain	Text length	Age group	Prior Knowledge		Outcome			Linguistic Modification								g	Var
							Content	Language	Recall	Apply	Transfer	1	2	3a	3b	4	5	6a	6b		
Follmer & Sperling	2018	1	305	BIO	1029	p-sec	+	0	X	X				X						0.07	0.013
Gasparinatou & Grigoriadou	2010	1	31	CSC	1500	p-sec	–	0	X					X	X	X				0.50	0.110
							–	0		X				X	X	X				0.31	0.113
							–	0			X			X	X	X				0.28	0.111
							–	0	X								X			0.47	0.110
							–	0		X							X			0.11	0.110
							–	0			X						X			0.20	0.111
		2	27	CSC	1500	p-sec	+	0	X					X	X	X				0.18	0.121
							+	0		X				X	X	X				–0.35	0.128
							+	0			X			X	X	X				–0.43	0.129
							+	0	X								X			0.16	0.121
							+	0		X							X			–0.13	0.126
							+	0			X						X			–0.21	0.126
							+	0	X					X	X	X				–0.31	0.053
							+	0	X								X			0.04	0.052
Gasparinatou & Grigoriadou	2011	1	65	CSC	900	p-sec	+	0		X				X	X	X				–0.49	0.056
							+	0		X							X			–0.06	0.054
							+	0			X			X	X	X				–0.51	0.056
							+	0			X			X	X	X		X		–0.34	0.057
		2	73	CSC	1425	p-sec	+	0	X					X	X	X				0.40	0.039
							+	0		X				X	X	X				–0.24	0.039
							+	0			X			X	X	X				–0.73	0.041
							–	0	X					X	X	X				0.78	0.059
							–	0		X				X	X	X				0.59	0.057
							–	0			X			X	X	X				0.85	0.060
Geva & Ryan	1985	1	13	MIX	250	sec1	0	–		X					X					–0.12	0.095
							0	–	X						X					–0.38	0.102
							0	–		X					X					–0.07	0.095
							0	–	X						X					0.04	0.094
							0	–		X					X					–0.70	0.121
							0	–	X						X					–0.17	0.096
		2	30	MIX	250	sec1	0	0		X					X					–0.19	0.045
							0	0	X						X					–0.28	0.046
							0	0		X					X					0.09	0.044
							0	0	X						X					0.01	0.044
							0	0		X					X					–0.43	0.049
							0	0	X						X					–0.25	0.046
		3	13	MIX	250	sec1	0	+		X					X					0.12	0.095
							0	+	X						X					–0.61	0.114
							0	+		X					X					0.48	0.107
							0	+	X						X					0.00	0.094
							0	+		X					X					–0.09	0.095
							0	+	X						X					–0.36	0.101
Hacker, Hübner, Müller, Nedkova, Osterland, & Wollenberger	1992	1	20	BIO	201	p-sec	0	0	X					X	X	X		X	X	–0.01	0.057
Hall, Basran, Paterson, Kowalski, Filik, & Maltby	2014	1	60	BIO	60	sec1	0	0		X				X	X	X				0.50	0.014
Hall, Kowalski, Paterson, Basran, Filik, & Maltby	2015	1	104	MIX	60	sec1	0	0		X				X	X	X				0.25	0.014
Hall, Maltby, Filik, & Paterson	2016	1	60	MIX	50	sec1	0	0		X				X						0.50	0.022
		2	71	MIX	50	sec1	0	0		X						X				–0.07	0.011
Härtig, Fraser, Bernholt, & Retelsdorf	2019	1	261	CHE	–99	sec1	0	0	X	X	X	X	X	X	X	X	X	X	X	–0.01	0.014
Ikeda & Kitagami Kamalski, Sanders, & Lentz	2013	1	52	MIX	330	p-sec	0	0	X	X				X	X	X	X			0.11	0.075
		1	25	BIO	500	p-sec	–	0		X				X	X		X			0.70	0.159
		2	54	BIO	500	p-sec	+	0		X				X	X	X				–0.20	0.072
		3	131	BIO	500	p-sec	0	0		X					X		X			0.21	0.030
		4	128	BIO	500	p-sec	–	0		X					X		X			0.34	0.016
		5	127	BIO	500	p-sec	+	0		X					X		X			–0.35	0.016

(continued on next page)

Table A.3 (continued)

Authors	Year	Group	N	Domain	Text length	Age group	Prior Knowledge		Outcome			Linguistic Modification						g	Var
							Content	Language	Recall	Apply	Transfer	1	2	3a	3b	4	5		
Lippmann, Schwartz, Jacobson, & Narciss	2019	1	124	MIX	250	p-sec	0	+	X								X	0.19	0.032
Maury, & Teisserenc	2005	1	89	MIX	62	p-sec	0	0	X					X				0.26	0.016
McNamara	2001	1	20	BIO	775	p-sec	–	0	X					X	X	X	X	1.33	0.228
						p-sec	–	0		X				X	X	X	X	0.66	0.194
						p-sec	+	0	X					X	X	X	X	–0.82	0.200
						p-sec	+	0		X				X	X	X	X	0.12	0.184
McNamara, Kintsch, Songer, & Kintsch	1996	1	56	BIO	868	secl	+	0	X					X	X	X		0.24	0.063
						secl	+	0		X				X	X	X		–0.35	0.071
						secl	+	0			X			X	X	X		–0.06	0.063
						secl	+	0	X								X	0.26	0.063
						secl	+	0		X							X	–0.47	0.071
						secl	+	0			X						X	–0.29	0.063
						secl	–	0	X					X	X	X		0.25	0.063
						secl	–	0		X				X	X	X		0.08	0.070
						secl	–	0			X			X	X	X		0.27	0.063
						secl	–	0	X								X	0.43	0.064
McNamara, Ozuru & Floyd	2011	1	33	BIO	507	pri	+	0		X				X	X	X	X	0.17	0.042
pri						+	0	X					X	X	X	X	0.25	0.039	
pri						0	+		X				X	X	X	X	–0.05	0.041	
pri						0	+	X					X	X	X	X	0.07	0.037	
pri						–	0		X				X	X	X	X	–0.17	0.042	
pri						–	0	X					X	X	X	X	–0.24	0.039	
pri						0	–		X				X	X	X	X	0.05	0.041	
pri						0	–	X					X	X	X	X	0.01	0.039	
Meneses, Escobar, & Véliz	2018	1	33	BIO	401	pri	0	–	X	X	X		X	X		X	0.74	0.129	
						pri	0	0	X	X	X		X	X		X	–0.05	0.051	
						pri	0	+	X	X	X		X	X		X	–0.19	0.078	
Meyer, Brandt & Bluth,	1980	1	32	MIX	204	secl	0	+	X					X	X	X	–0.11	0.107	
						secl	0	+	X					X	X	X	–0.31	0.108	
						secl	0	0	X					X	X	X	0.33	0.103	
						secl	0	0	X					X	X	X	0.03	0.100	
		3	34	MIX	204	secl	0	–	X					X	X	X	0.36	0.106	
						secl	0	–	X					X	X	X	–0.34	0.111	
						secl	0	–	X					X			–1.10	0.068	
						p-sec	0	0	X				X			–0.49	0.047		
Millis, Graesser, & Haberlandt	1993	1	32	MIX	55	p-sec	0	0	X					X			–0.25	0.043	
						p-sec	0	0	X					X			–0.67	0.035	
						p-sec	0	0	X					X			–0.09	0.028	
						p-sec	0	0	X					X			–0.33	0.030	
		3	30	MIX	55	p-sec	0	0	X					X			–0.35	0.044	
						p-sec	0	0	X					X			1.00	0.065	
						p-sec	–	0		X							X	1.58	0.116
						p-sec	–	0	X								X	0.20	0.088
O'Reilly & McNamara	2007	1	39	BIO	776	p-sec	–	–	X					X	X	X	X	–0.09	0.099
						p-sec	–	–		X				X	X	X	X	0.89	0.109
						p-sec	–	+	X					X	X	X	X	0.08	0.119
						p-sec	–	+		X				X	X	X	X	0.83	0.130
		3	30	BIO	50	p-sec	+	–	X					X	X	X	X	–0.76	0.136
						p-sec	+	–		X				X	X	X	X	–0.30	0.128
						p-sec	+	+	X					X	X	X	X	0.64	0.097
						p-sec	0	+		X				X	X	X	X	0.00	0.092
Ortegren, Serra, & England	2015	1	80	PHY	606	p-sec	0	0	X							X	0.40	0.045	
						p-sec	0	0						X			X	0.20	0.049
						p-sec	0	0	X							X	0.52	0.047	
						p-sec	+	0		X				X			0.00	0.049	
Ozuru, Briner, Best, & McNamara	2010	1	78	PSY	402	p-sec	0	0	X					X	X		–0.44	0.052	
						p-sec	0	0		X				X	X			–0.05	0.045
Ozuru, Dempsey, & McNamara	2009	1	23	BIO	803	p-sec	+	+	X					X	X	X	X	0.57	0.066
						p-sec	+	+		X				X	X	X	X	–0.02	0.029
						p-sec	+	–	X					X	X	X	X	–0.38	0.037
						p-sec	+	–		X				X	X	X	X	–0.02	0.017
						p-sec	–	+	X					X	X	X	X	0.21	0.023

(continued on next page)

Table A.3 (continued)

Authors	Year	Group	N	Domain	Text length	Age group	Prior Knowledge		Outcome			Linguistic Modification						g	Var				
							Content	Language	Recall	Apply	Transfer	1	2	3a	3b	4	5			6a	6b		
Plath	2020	4	46	BIO	803	p-sec	–	+		X				X	X	X	X		0.01	0.011			
						p-sec	–	–	X					X	X	X	X		0.13	0.030			
						p-sec	–	–		X				X	X	X	X		–0.10	0.027			
						sec1	0	0		X		X	X						0.10	0.065			
						sec1	0	0		X			X	X					–0.06	0.092			
Sánchez, García, & Bustos	2017	1	192	GEO	378	sec1	–	0	X								X		0.45	0.021			
Schneider	2018	1	213	BIO	1000	sec1	0	0		X			X	X		X	X		0.51	0.019			
Snyder	2012	1	40	MIX	91	pri	–	0	X							X	X		–0.13	0.088			
						pri	–	0	X							X	X		–0.07	0.087			
Spyridakis & Standal	1987	1	296	MIX	540	p-sec	+	+	X									X		0.01	0.013		
						p-sec	+	+		X								X		0.08	0.014		
						p-sec	+	+	X									X		0.01	0.014		
						p-sec	+	+		X								X		0.38	0.014		
						p-sec	+	+	X							X				0.42	0.014		
Stiller & Jedlicka	2010	1	65	BIO	1126	p-sec	+	+		X						X			0.26	0.014			
						sec2	+	0			X								X		–0.72	0.127	
						sec2	+	0		X									X		0.11	0.119	
						sec2	–	0			X								X		0.51	0.127	
						sec2	–	0		X									X		0.83	0.133	
Vidal-Abarca & Sanjose	1998	1 + 2	70	BIO	1126	sec2	0	0	X									X		0.71	0.064		
						sec1	0	0	X												0.70	0.056	
						sec1	0	0		X									X		0.76	0.063	
						sec1	0	0			X								X		0.46	0.060	
						sec1	0	0	X							X	X	X	X		0.60	0.056	
Walkington, Clinton, & Shivraj	2019	1	246	MAT	50	sec1	0	0		X										0.99	0.066		
						sec1	0	0			X							X	X	X	X	0.92	0.065
						sec2	0	0		X					X						0.11	0.013	
Wenger & Spyridakis	1993	1	22	GEO	1034	p-sec	0	0			X	X								0.67	0.187		
						p-sec	0	0	X			X									0.40	0.162	
						p-sec	0	0			X	X	X			X	X				0.12	0.186	
						p-sec	0	0	X			X	X			X	X				0.47	0.176	
						p-sec	–	0			X	X									1.12	0.247	
Williams	1986	2	130	BIO	2000	p-sec	–	0	X			X							–0.25	0.193			
						p-sec	–	0			X	X	X			X	X				–0.16	0.198	
						p-sec	–	0	X			X	X			X	X				0.33	0.182	
						pri	0	+				X				X					0.31	0.031	
						pri	0	0				X				X					0.11	0.037	
Wittwer & Ihme	2014	1	55	PHY	144	pri	0	–				X				X			0.40	0.047			
						p-sec	0	0		X				X							0.04	0.021	
Yaros	2006	1	234	BIO	709	p-sec	0	0		X							X		0.33	0.023			
						p-sec	+	0	X							X	X				0.30	0.017	
Yaros	2011	1	298	BIO	709	p-sec	+	0		X							X	X		0.51	0.018		
						p-sec	+	0	X	X						X	X				0.30	0.012	

Note. BIO = biology, CHE = chemistry, CSC = computer sciences, GEO = geography, MAT = mathematics, MIX = mixed science, PHY = physics, PSY = psychology. pri = primary education, sec1 = lower secondary education, sec2 = upper secondary education, p-sec = post-secondary education. – = low prior knowledge, 0 = regular prior knowledge, + = high prior knowledge. 1 = reducing word-level complexity, 2 = reducing syntactic complexity, 3a = increasing local cohesion/overlap, 3b = increasing local cohesion/connectives, 4 = increasing global cohesion, 5 = increasing local clarity and elaboration, 6a = increasing global clarity and elaboration/summaries, 6b = increasing global clarity and elaboration/personalization.

Data availability

Data will be made available on request.

References

- Studies included in the meta-analysis are marked with an asterisk
- Alexander, P. A., Kulikowich, J. M., & Schulze, S. K. (1994). How subject-matter knowledge affects recall and interest. *American Educational Research Journal*, 31(2), 313–337. <https://doi.org/10.3102/00028312031002313>
- * Arya, D. J., Hiebert, E. H., & Pearson, P. D. (2011). The effects of syntactic and lexical complexity on the comprehension of elementary science texts. *International Electronic Journal of Environmental Education*, 4(1), 107–125.
- Best, R., Rowe, M., Ozuru, Y., & McNamara, D. (2005). Deep-level comprehension of science texts: The role of the reader and the text. *Topics in Language Disorders*, 25(1), 65–83. <https://doi.org/10.1097/00011363-200501000-00007>
- Bohn-Gettler, C. M., & Kendeou, P. (2014). The interplay of reader goals, working memory, and text structure during reading. *Contemporary Educational Psychology*, 39(3), 206–219. <https://doi.org/10.1016/j.cedpsych.2014.05.003>

- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. Wiley.
- * Boscolo, P., & Mason, L. (2003). Topic knowledge, text coherence, and interest: How they interact in learning from instructional texts. *The Journal of Experimental Education*, 71(2), 126–148.
- van den Broek, P. (2010). Using texts in science education: Cognitive processes and knowledge representation. *Science*, 328(5977), 453–456. <https://doi.org/10.1126/science.1182594>
- van den Broek, P., Virtue, S., Everson, M. G., Tzeng, Y., & Sung, Y.-C. (2002). Comprehension and memory of science texts: Inferential processes and the construction of a mental representation. In J. Otero, J. A. León, & A. C. Graesser (Eds.), *The psychology of science text comprehension* (pp. 131–154). <https://doi.org/10.4324/9781410612434>. Routledge.
- * Burton, R. (2014). Readability, logodiversity, and the effectiveness of college science textbooks. *Bioscene*, 40, 3–10.
- Butterfuss, R., & Kendeou, P. (2017). The role of executive functions in reading comprehension. *Educational Psychology Review*, 30, 801–826. <https://doi.org/10.1007/s10648-017-9422-6>
- Cheung, M. W. L. (2014). Modeling dependent effect sizes with three-level meta-analyses: A structural equation modeling approach. *Psychological Methods*, 19(2), 211–229. <https://doi.org/10.1037/a0032968>
- Cordova, D. I., & Lepper, M. R. (1996). Intrinsic motivation and the process of learning: Beneficial effects of contextualization, personalization, and choice. *Journal of Educational Psychology*, 88(4), 715–730. <https://doi.org/10.1037/0022-0663.88.4.715>
- Coté, N., Goldman, S., & Saul, E. (1998). Students making sense of informational text: Relations between processing and representation. *Discourse Processes*, 25(1), 1–53. <https://doi.org/10.1080/01638539809545019>
- Cromley, J. G., & Azevedo, R. (2007). Testing and refining the direct and inferential mediation model of reading comprehension. *Journal of Educational Psychology*, 99(2), 311–325. <https://doi.org/10.1037/0022-0663.99.2.311>
- Cromley, J., Snyder-Hogan, L., & Dubas, U. (2010). Reading comprehension of scientific text: A domain-specific test of the direct and inferential mediation model of reading comprehension. *Journal of Educational Psychology*, 102, 687–700. <https://doi.org/10.1037/a0019452>
- Cruz Neri, N., & Retelsdorf, J. (2022). The role of linguistic features in science and math comprehension and performance: A systematic review and desiderata for future research. *Educational Research Review*, 36, Article 100460. <https://doi.org/10.1016/j.edurev.2022.100460>
- Deese, J. (1962). *On the structure of associative meaning*. American Psychological Association. <https://doi.org/10.1037/h0045842>
- * Degand, L., Lefevre, N., & Bestgen, Y. (1999). The impact of connectives and anaphoric expressions on expository discourse comprehension. *Document Design*, 1(1), 39–51. <https://doi.org/10.1075/dd.1.1.06deg>
- * Degand, L., & Sanders, T. (2002). The impact of relational markers on expository text comprehension in L1 and L2. *Reading and Writing: An Interdisciplinary Journal*, 15(7–8), 739–757. <https://doi.org/10.1023/A:1020932715838>
- * Deppert, A. (2001). *Verstehen und Verständlichkeit*. Deutscher Universitäts-Verlag.
- van Dijk, T. A., & Kintsch, W. (1983). *Strategies of discourse comprehension*. Academic Press.
- * Dutke, S., Grefe, A. C., & Leopold, C. (2016). Learning from scientific texts: Personalizing the text increases transfer performance and task involvement. *European Journal of Psychology of Education*, 31(4), 499–513. <https://doi.org/10.1007/s10212-015-0281-6>
- Fang, Z., Lamme, L., Pringle, R., Patrick, J., Sanders, J., Zmach, C., Charbonnet, S., & Henkel, M. (2008). Integrating reading into middle school science: What we did, found and learned. *International Journal of Education*, 30(15), 2067–2089. <https://doi.org/10.1080/09500690701644266>
- Fang, Z., Schleppegrell, M. J., & Cox, B. E. (2006). Understanding the language demands of schooling: Nouns in academic registers. *Journal of Literacy Research*, 38(3), 247–273. https://doi.org/10.1207/s15548430jlr3803_1
- Fernández-Castilla, B., Maes, M., Declercq, L., Jamshidi, L., Beretvas, S. N., Onghena, P., & Van den Noortgate, W. (2019). A demonstration and evaluation of the use of cross-classified random-effects models for meta-analysis. *Behavior Research Methods*, 51(3), 1286–1304. <https://doi.org/10.3758/s13428-018-1063-2>
- Flesch, R. (1948). A new readability yardstick. *Journal of Applied Psychology*, 32(3), 221–233. <https://doi.org/10.1037/h0057532>
- Follmer, D. J., Fang, S.-Y., Clariana, R. B., Meyer, J. F., & Li, P. (2018). What predicts adult readers' understanding of STEM texts? *Reading and Writing*, 31(1), 185–214. <https://doi.org/10.1007/s11145-017-9781-x>
- * Follmer, D. J., & Sperling, R. A. (2018). Interactions between reader and text: Contributions of cognitive processes, strategy use, and text cohesion to comprehension of expository science text. *Learning and Individual Differences*, 67, 177–187. <https://doi.org/10.1016/j.lindif.2018.08.005>
- * Gasparinatou, A., & Grigoriadou, M. (2010). Learning from texts in computer science. *The International Journal of Learning*, 17(1).
- * Gasparinatou, A., & Grigoriadou, M. (2011). Supporting students' learning in the domain of computer science. *Computer Science Education*, 21(1), 1–28. <https://doi.org/10.1080/089933408.2010.509909>
- * Gasparinatou, A., & Grigoriadou, M. (2013). Exploring the effect of background knowledge and text cohesion on learning from texts in computer science. *Educational Psychology*, 33(6), 645–670. <https://doi.org/10.1080/01443410.2013.790309>
- * Geva, E., & Ryan, E. B. (1985). Use of conjunctions in expository texts by skilled and less skilled readers. *Journal of Reading Behavior*, 17(4), 331–346. <https://doi.org/10.1080/10862968509547549>
- Ginns, P., Martin, A. J., & Marsh, H. W. (2013). Designing instructional text in a conversational style: A meta-analysis. *Educational Psychology Review*, 25(4), 445–472. <https://doi.org/10.1007/s10648-013-9228-0>
- Goldman, S. R., & Bisanz, G. L. (2002). Toward a functional analysis of scientific genres: Implications for understanding and learning processes. In J. Otero, J. A. León, & A. C. Graesser (Eds.), *The psychology of science text comprehension* (pp. 19–50). <https://doi.org/10.4324/9781410612434>. Routledge.
- Graesser, A. C., León, J. A., & Otero, J. (2002). Introduction to the psychology of science text comprehension. In J. Otero, J. A. León, & A. C. Graesser (Eds.), *The psychology of science text comprehension* (pp. 1–15). <https://doi.org/10.4324/9781410612434>. Routledge.
- Graesser, A. C., McNamara, D. S., & Kulikowich, J. M. (2011). Coh-Matrix: Providing multilevel analyses of text characteristics. *Educational Researcher*, 40(5), 223–234. <https://doi.org/10.3102/0013189X11413260>
- Guo, D., Zhang, S., Wright, K. L., & McTigue, E. M. (2020). Do you get the picture? A meta-analysis of the effect of graphics on reading comprehension. *AERA Open*, 6(1), Article 2332858420901696. <https://doi.org/10.1177/2332858420901696>
- * Hacker, W., Hübner, I., Müller, H., Nedkova, S., Osterland, D., & Wollenberger, E. (1992). Profitiert nur der Bessere? Zum Einfluss von Text- und arbeitsgedächtnismerkmalen auf das Textverstehen. *Sprache & Kognition*, 11(4), 208–222.
- * Hall, S., Basran, J., Paterson, K. B., Kowalski, R. K., Filik, R., & Maltby, J. (2014). Individual differences in the effectiveness of text cohesion for science text comprehension. *Learning and Individual Differences*, 29, 74–80. <https://doi.org/10.1016/j.lindif.2013.10.014>
- * Hall, S., Kowalski, R., Paterson, K., Basran, J., Filik, R., & Maltby, J. (2015). Local text cohesion, reading ability and individual science aspirations: Key factors influencing comprehension in science classes. *British Educational Research Journal*, 41(1), 122–142. <https://doi.org/10.1002/berj.3134>
- * Hall, S. S., Maltby, J., Filik, R., & Paterson, K. B. (2016). Key skills for science learning: The importance of text cohesion and reading ability. *Educational Psychology*, 36(2), 191–215. <https://doi.org/10.1080/01443410.2014.926313>
- Hand, B., Yore, L. D., Jagger, S., & Prain, V. (2010). Connecting research in science literacy and classroom practice: A review of science teaching journals in Australia, the UK and the United States, 1998–2008. *Studies in Science Education*, 46(1), 45–68. <https://doi.org/10.1080/03057260903562342>
- * Härtig, H., Fraser, N., Bernholt, S., & Retelsdorf, J. (2019). Kann man Sachtexte vereinfachen? – Ergebnisse einer Generalisierungsstudie zum Textverständnis. *Zeitschrift für Didaktik der Naturwissenschaften*, 25(1), 273–287. <https://doi.org/10.1007/s40573-019-00105-7>
- Hedges, L. V. (1981). Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2), 107–128. <https://doi.org/10.2307/1164588>
- Higgins, J. P., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11), 1539–1558. <https://doi.org/10.1002/sim.1186>
- * Ikeda, K., & Kitagami, S. (2013). The interactive effect of working memory and text difficulty on metacomprehension accuracy. *Journal of Cognitive Psychology*, 25(1), 94–106. <https://doi.org/10.1080/20445911.2012.748028>
- Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, 38(1), 23–31. https://doi.org/10.1207/S15326985EP3801_4

- * Kamalski, J., Sanders, T., & Lentz, L. (2008). Coherence marking, prior knowledge, and comprehension of informative and persuasive texts: Sorting things out. *Discourse Processes*, 45(4–5), 323–345. <https://doi.org/10.1080/01638530802145486>.
- Kincaid, J. P., Fishburne, R. P., Rogers, R. L., & Chissom, B. S. (1975). *Derivation of new readability formulas (automated readability index, fog count and Flesch reading ease formula) for navy enlisted personnel*. Naval Technical Training Command Millington TN Research Branch.
- Kintsch, W., & van Dijk, T. A. (1978). Toward a model of text comprehension and production. *Psychological Review*, 85, 363–394. <https://doi.org/10.1037/0033-295X.85.5.363>
- Knapp, G., & Hartung, J. (2003). Improved tests for a random effects meta-regression with a single covariate. *Statistics in Medicine*, 22(17), 2693–2710. <https://doi.org/10.1002/sim.1482>
- Leutner, D., Leopold, C., & Sumfleth, E. (2009). Cognitive load and science text comprehension: Effects of drawing and mentally imagining text content. *Computers in Human Behavior*, 25(2), 284–289. <https://doi.org/10.1016/j.chb.2008.12.010>
- * Lippmann, M., Schwartz, N. H., Jacobson, N. G., & Narciss, S. (2019). The concreteness of titles affects metacognition and study motivation. *Instructional Science*, 47(3), 257–277. <https://doi.org/10.1007/s11251-018-9478-9>.
- Lipsey, M. W., & Wilson, D. B. (2000). *Practical meta-analysis*. Sage.
- * Maury, P., & Teisserenc, A. (2005). The role of connectives in science text comprehension and memory. *Language & Cognitive Processes*, 20(3), 489–512. <https://doi.org/10.1080/01690960444000151>.
- Mayer, R., Heiser, J., & Lonn, S. (2001). Cognitive constraints on multimedia learning: When presenting more material results in less understanding. *Journal of Educational Psychology*, 93, 187–198. <https://doi.org/10.1037/0022-0663.93.1.187>
- * McNamara, D. S. (2001). Reading both high-coherence and low-coherence texts: Effects of text sequence and prior knowledge. *Canadian Journal of Experimental Psychology/Revue canadienne de psychologie experimentale*, 55(1), 51–62. <https://doi.org/10.1037/h0087352>.
- McNamara, D. S., & Kintsch, W. (1996). Learning from texts: Effects of prior knowledge and text coherence. *Discourse Processes*, 22(3), 247–288. <https://doi.org/10.1080/01638539609544975>
- * McNamara, D. S., Kintsch, E., Songer, N. B., & Kintsch, W. (1996). Are good texts always better? Interactions of text coherence, background knowledge, and levels of understanding in learning from text. *Cognition and Instruction*, 14(1), 1–43. https://doi.org/10.1207/s1532690xci1401_1.
- McNamara, D. S., Louwerse, M. M., McCarthy, P. M., & Graesser, A. C. (2010). Coh-Matrix: Capturing linguistic features of cohesion. *Discourse Processes*, 47(4), 292–330. <https://doi.org/10.1080/01638530902959943>
- McNamara, D. S., & Magliano, J. (2009). Toward a comprehensive model of comprehension. *Psychology of Learning and Motivation*, 51, 297–384. [https://doi.org/10.1016/S0079-7421\(09\)51009-2](https://doi.org/10.1016/S0079-7421(09)51009-2)
- * McNamara, D. S., Ozuru, Y., & Floyd, R. G. (2011). Comprehension challenges in the fourth grade: The roles of text cohesion, text genre, and readers' prior knowledge. *International Electronic Journal of Environmental Education*, 4(1), 229–257.
- * Meneses, A., Escobar, J.-P., & Véliz, S. (2018). The effects of multimodal texts on science reading comprehension in Chilean fifth-graders: Text scaffolding and comprehension skills. *International Journal of Science Education*, 40(18), 2226–2244. <https://doi.org/10.1080/09500693.2018.1527472>.
- van Merriënboer, J. J. G., & Sweller, J. (2005). Cognitive load theory and complex learning: Recent developments and future directions. *Educational Psychology Review*, 17(2), 147–177. <https://doi.org/10.1007/s10648-005-3951-0>
- * Meyer, B. J. F., Brandt, D. M., & Bluth, G. J. (1980). Use of top-level structure in text: Key for reading comprehension of ninth-grade students. *Reading Research Quarterly*, 16(1), 72–103. <https://doi.org/10.2307/747349>.
- Mikk, J. (2008). Sentence length for revealing the cognitive load reversal effect in text comprehension. *Educational Studies*, 34(2), 119–127. <https://doi.org/10.1080/03055690701811164>
- * Millis, K. K., Graesser, A. C., & Haberlandt, K. F. (1993). The impact of connectives on the memory for expository texts. *Applied Cognitive Psychology*, 7(4), 317–339. <https://doi.org/10.1002/acp.2350070406>.
- * Moreno, R., & Mayer, R. E. (2000). Engaging students in active learning: The case for personalized multimedia messages. *Journal of Educational Psychology*, 92(4), 724–733. <https://doi.org/10.1037/0022-0663.92.4.724>.
- * O'Reilly, T., & McNamara, D. S. (2007). Reversing the reverse cohesion effect: Good texts can be better for strategic, high-knowledge readers. *Discourse Processes*, 43(2), 121–152. <https://doi.org/10.1080/01638530709336895>.
- * Ortegren, F. R., Serra, M. J., & England, B. D. (2015). Examining competing hypotheses for the effects of diagrams on recall for text. *Memory & Cognition*, 43(1), 70–84. <https://doi.org/10.3758/s13421-014-0429-7>.
- * Ozuru, Y., Briner, S., Best, R., & McNamara, D. S. (2010). Contributions of self-explanation to comprehension of high- and low-cohesion texts. *Discourse Processes*, 47(8), 641–667. <https://doi.org/10.1080/01638531003628809>.
- * Ozuru, Y., Dempsey, K., & McNamara, D. S. (2009). Prior knowledge, reading skill, and text cohesion in the comprehension of science texts. *Learning and Instruction*, 19(3), 228–242. <https://doi.org/10.1016/j.learninstruc.2008.04.003>.
- Paas, F., Renkl, A., & Sweller, J. (2003). Cognitive load theory and instructional design: Recent developments. *Educational Psychologist*, 38(1), 1–4. https://doi.org/10.1207/s15326985EP3801_1
- * Plath, J. (2020). Verstehensprozesse bei der Bearbeitung realitätsbezogener Mathematikaufgaben: Klassische Textaufgaben vs. Zeitungstexte. *Journal für Mathematik-Didaktik*, 41(2), 237–266. <https://doi.org/10.1007/s13138-019-00148-w>.
- Retelsdorf, J., Köller, O., & Möller, J. (2011). On the effects of motivation on reading performance growth in secondary school. *Learning and Instruction*, 21(4), 550–559. <https://doi.org/10.1016/j.learninstruc.2010.11.001>
- Rodgers, M. A., & Pustejovsky, J. E. (2020). Evaluating meta-analytic methods to detect selective reporting in the presence of dependent effect sizes. *Psychological Methods*, 26(2), 141–160. <https://doi.org/10.1037/met0000300>
- * Sánchez, E., García, J. R., & Bustos, A. (2017). Does rhetorical competence moderate the effect of rhetorical devices on the comprehension of expository texts beyond general comprehension skills? *Reading and Writing*, 30(3), 439–462. <https://doi.org/10.1007/s11145-016-9684-2>.
- Schleppegrell, M. J. (2001). Linguistic features of the language of schooling. *Linguistics and Education*, 12(4), 431–459. [https://doi.org/10.1016/S0898-5898\(01\)00073-0](https://doi.org/10.1016/S0898-5898(01)00073-0)
- Schleppegrell, M. J. (2007). The linguistic challenges of mathematics teaching and learning. *Reading & Writing Quarterly*, 23, 139–159. <https://doi.org/10.1080/10573560601158461>
- * Schneider, H., Dittmar, M., Gilg, E., & Schmellentin, C. (2018). Textseitige Maßnahmen zur Unterstützung des Leseverstehens im Biologieunterricht. *Didaktik Deutsch*, 23(45), 94–116.
- Schnotz, W., & Bannert, M. (2003). Construction and interference in learning from multiple representation. *Learning and Instruction*, 13(2), 141–156. [https://doi.org/10.1016/S0959-4752\(02\)00017-8](https://doi.org/10.1016/S0959-4752(02)00017-8)
- Snow, C. (2002). *Reading for understanding: Toward an R&D program in reading comprehension*. RAND.
- Snow, C., & Uccelli, P. (2009). The challenge of academic language. In D. R. Olson, & N. Torrance (Eds.), *The cambridge handbook of literacy* (pp. 112–133). Cambridge University Press. <https://doi.org/10.1017/CBO9780511609664.008>.
- * Snyder, A. E. (2012). *The effects of graphic organizers and content familiarity on second graders' comprehension of cause/effect text* [doctoral thesis]. New York, NY: Columbia University.
- * Spyridakis, J. H., & Standal, T. C. (1987). Signals in expository prose: Effects on reading comprehension. *Reading Research Quarterly*, 22(3), 285–298. <https://doi.org/10.2307/747969>.
- * Stiller, K., & Jedlicka, R. (2010). A kind of expertise reversal effect: Personalisation effect can depend on domain-specific prior knowledge. *Australasian Journal of Educational Technology*, 26, 133–149. <https://doi.org/10.14742/ajet.1107>.
- Sundararajan, N., & Adesope, O. (2020). Keep it coherent: A meta-analysis of the seductive details effect. *Educational Psychology Review*, 32(3), 707–734. <https://doi.org/10.1007/s10648-020-09522-4>

- Swart, E. K., Nielen, T. M. J., & Sikkema - de Jong, M. T. (2019). Supporting learning from text: A meta-analysis on the timing and content of effective feedback. *Educational Research Review*, 28, Article 100296. <https://doi.org/10.1016/j.edurev.2019.100296>
- Sweller, J. (1994). Cognitive load theory, learning difficulty, and instructional design. *Learning and Instruction*, 4(4), 295–312. [https://doi.org/10.1016/0959-4752\(94\)90003-5](https://doi.org/10.1016/0959-4752(94)90003-5)
- Sweller, J., van Merriënboer, J. J. G., & Paas, F. G. W. C. (1998). Cognitive architecture and instructional design. *Educational Psychology Review*, 10(3), 251–296. <https://doi.org/10.1023/A:1022193728205>
- Sweller, J., van Merriënboer, & Paas, F. (2019). Cognitive Architecture and Instructional Design: 20 Years Later. *Educational Psychology Review*, 31(2), 261–292. <https://doi.org/10.1007/s10648-019-09465-5>
- Tippett, C. D. (2010). Refutation text in science education: A review of two decades of research. *International Journal of Science and Mathematics Education*, 8, 951–970. <https://doi.org/10.1007/s10763-010-9203-x>
- * Vidal-Abarca, E., & Sanjose, V. (1998). Levels of comprehension of scientific prose: The role of text variables. *Learning and Instruction*, 8(3), 215–233.
- Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36(3), 1–48. <https://doi.org/10.18637/jss.v036.i03>
- Viechtbauer, W. (2022). I^2 for multilevel and multivariate models. Retrieved 18.03. from https://www.metafor-project.org/doku.php/tips:i2_multilevel_multivariate.
- * Walkington, C., Clinton, V., & Shivraj, P. (2017). How readability factors are differentially associated with performance for students of different backgrounds when solving mathematics word problems. *American Educational Research Journal*, 55(2), 362–414. <https://doi.org/10.3102/0002831217737028>.
- Weaver, C. A., III, & Kintsch, W. (1991). Expository text. In *Handbook of reading research* (Vol. 2, pp. 230–245). Lawrence Erlbaum Associates, Inc.
- * Wenger, M. J., & Spyridakis, J. H. (1993). Reduced text structure at two text levels: Impacts on the performance of technical readers. *Journal of Technical Writing and Communication*, 23(4), 333–352. <https://doi.org/10.2190/GWCQ-84CR-7DVB-RCNN>.
- * Williams, D. L. (1968). Rewritten science materials and reading comprehension. *The Journal of Educational Research*, 61(5), 204–206.
- * Wittwer, J., & Ihme, N. (2014). Reading skill moderates the impact of semantic similarity and causal specificity on the coherence of explanations. *Discourse Processes*, 51(1–2), 143–166. <https://doi.org/10.1080/0163853X.2013.855577>.
- Wittwer, J., & Renkl, A. (2008). Why instructional explanations often do not work: A framework for understanding the effectiveness of instructional explanations. *Educational Psychologist*, 43(1), 49–64. <https://doi.org/10.1080/00461520701756420>
- * Yaros, R. A. (2006). Is it the medium or the message? Structuring complex news to enhance engagement and situational understanding by nonexperts. *Communication Research*, 33(4), 285–309. <https://doi.org/10.1177/0093650206289154>.
- * Yaros, R. A. (2011). Effects of text and hypertext structures on user interest and understanding of science and technology. *Science Communication*, 33(3), 275–308. <https://doi.org/10.1177/1075547010386803>.